Sayısal Görüntü İşleme

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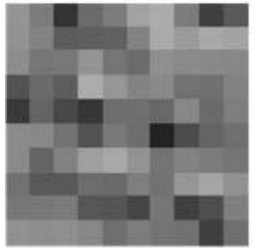
What is an Image?

- 2-dimensional matrix of Intensity (gray or color) values

Set of Intensity values

Image coordinates are integers

$$I(u, v) \in \mathbb{P} \quad \text{and} \quad u, v \in \mathbb{N}.$$



$F(x, y)$

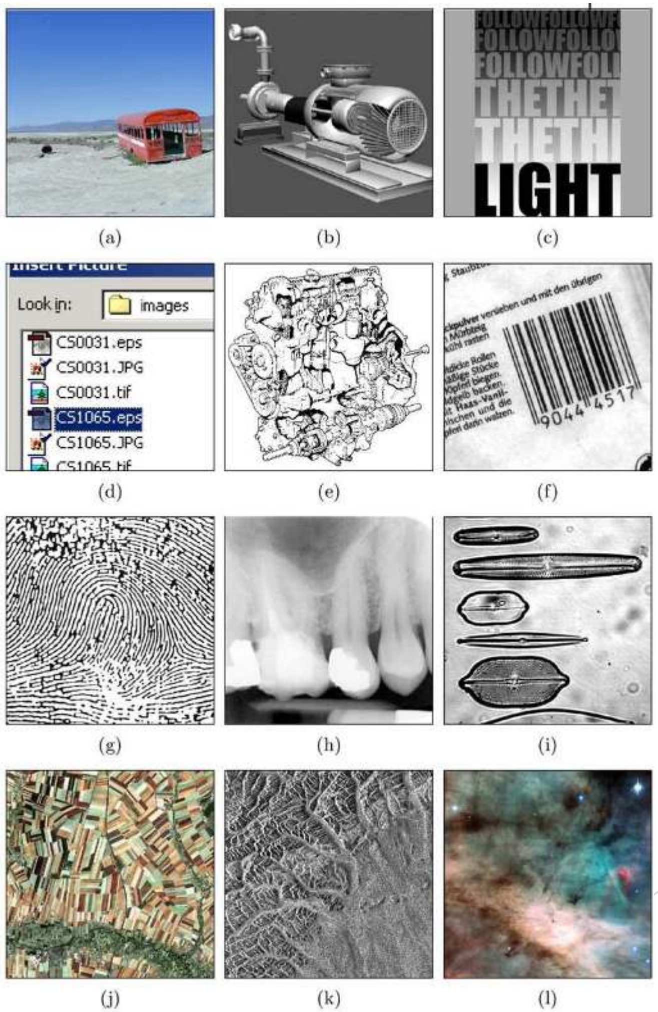
$\rightarrow$

$$\begin{array}{|ccccccccccc|}
148 & 123 & 52 & 107 & 123 & 162 & 172 & 123 & 64 & 89 & \cdots \\
147 & 130 & 92 & 95 & 98 & 130 & 171 & 155 & 169 & 163 & \cdots \\
141 & 118 & 121 & 148 & 117 & 107 & 144 & 137 & 136 & 134 & \cdots \\
82 & 106 & 93 & 172 & 149 & 131 & 138 & 114 & 113 & 129 & \cdots \\
57 & 101 & 72 & 54 & 109 & 111 & 104 & 135 & 106 & 125 & \cdots \\
138 & 135 & 114 & 82 & 121 & 110 & 34 & 76 & 101 & 111 & \cdots \\
138 & 102 & 128 & 159 & 168 & 147 & 116 & 129 & 124 & 117 & \cdots \\
113 & 89 & 89 & 109 & 106 & 126 & 114 & 150 & 164 & 145 & \cdots \\
120 & 121 & 123 & 87 & 85 & 70 & 119 & 64 & 79 & 127 & \cdots \\
145 & 141 & 143 & 134 & 111 & 124 & 117 & 113 & 64 & 112 & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots &
\end{array}$$

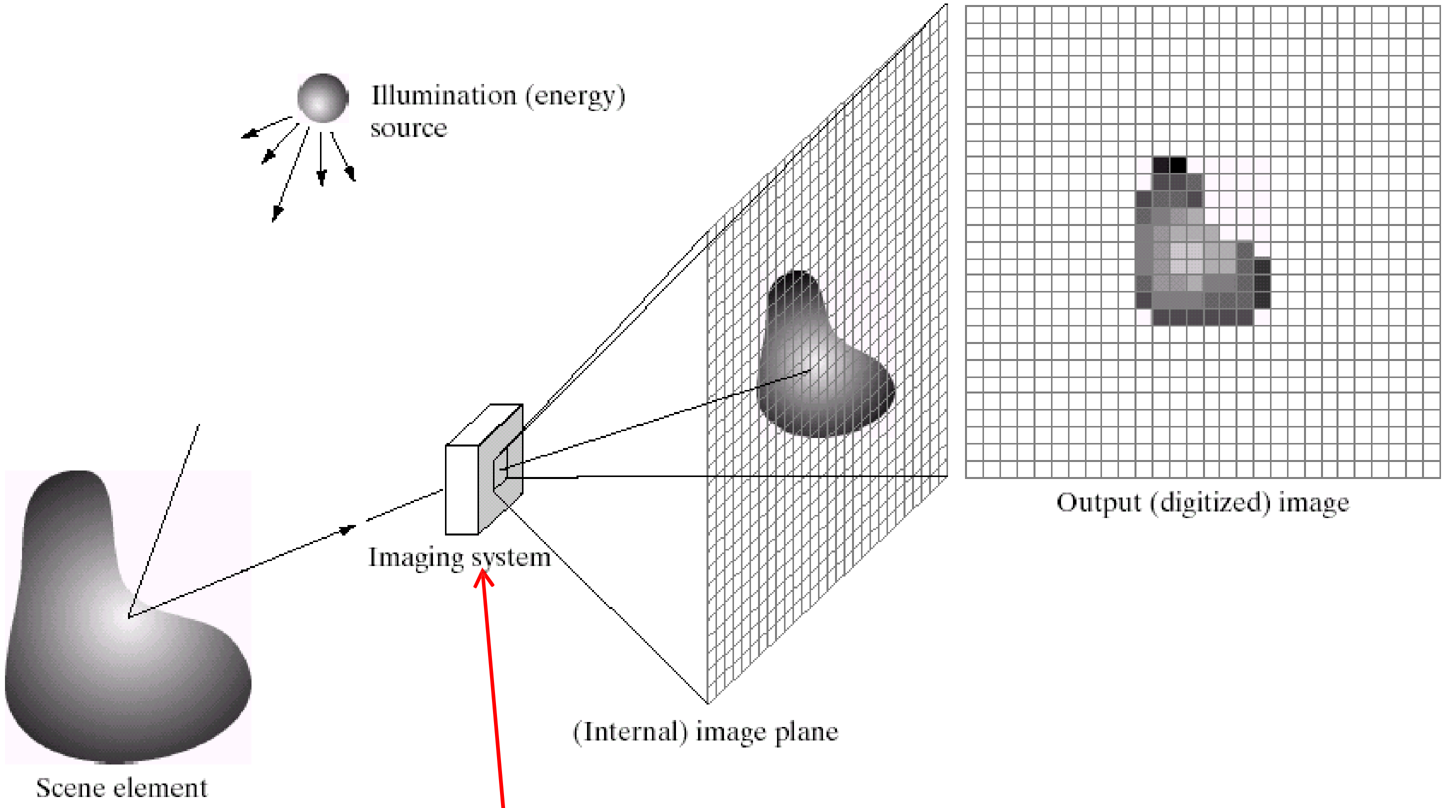
$I(u, v)$

Example of Digital Images

a) Natural landscape
b) Synthetically generated scene
c) Poster graphic
d) Computer screenshot
e) Black and white illustration
f) Barcode
g) Fingerprint
h) X-ray
i) Microscope slide
j) Satellite Image
k) Radar image
l) Astronomical object



Imaging System



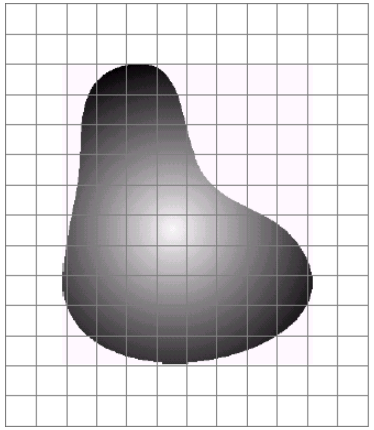
Example: a camera
Converts light to image

Credits: Gonzales and Woods

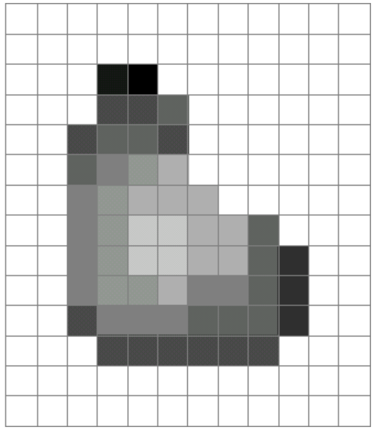
Digital Image?

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

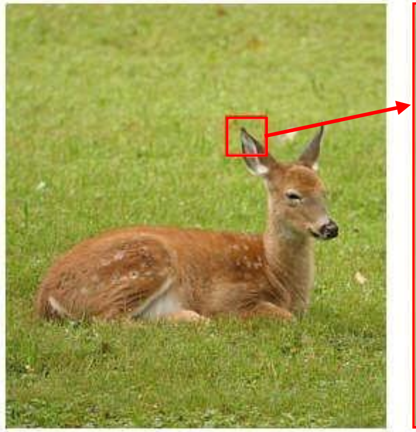
- Remember: *digitization* causes a digital image to become an *approximation* of a real scene



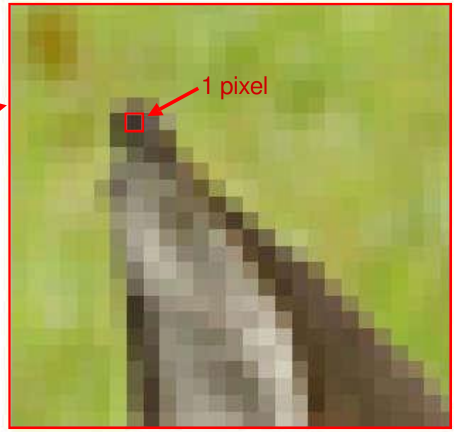
Real image



Digital Image
(an approximation)



Real image



Digital Image
(an approximation)

Digital Image

- Common image formats include:
  - 1 values per point/pixel (B&W or Grayscale)
  - 3 values per point/pixel (Red, Green, and Blue)
  - 4 values per point/pixel (Red, Green, Blue, + “Alpha” or Opacity)

Grayscale

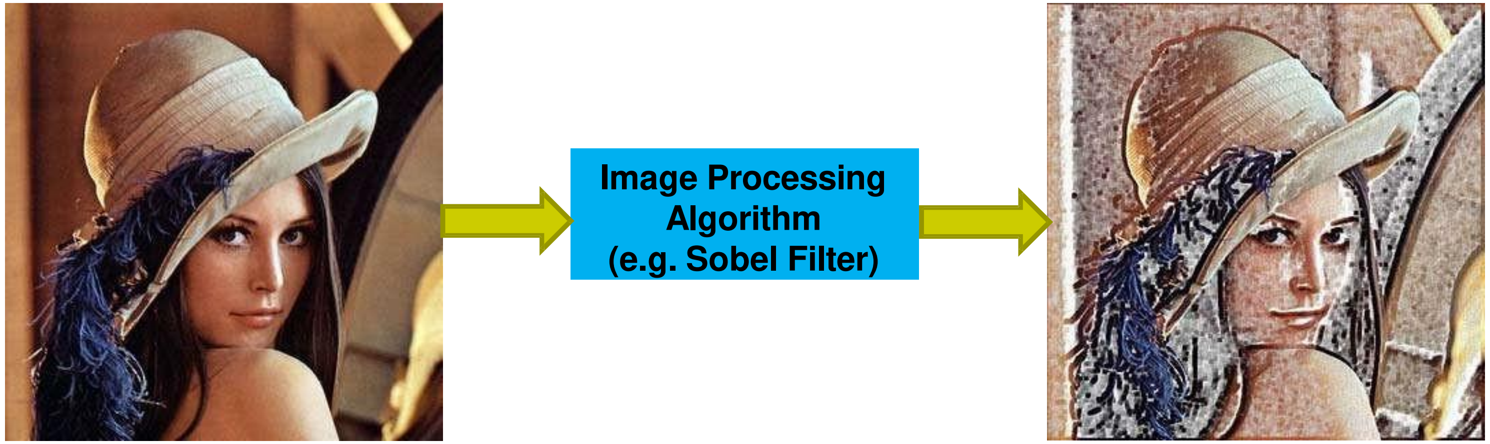
RGB

RGBA

- We will start with gray-scale images, extend to color later

What is image Processing?

- Algorithms that alter an input image to create new image
- Input is image, output is image

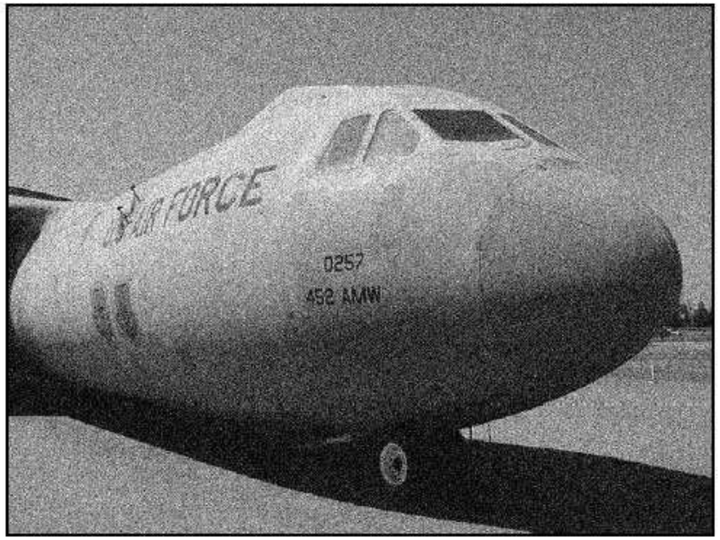


Original Image *Processed Image*

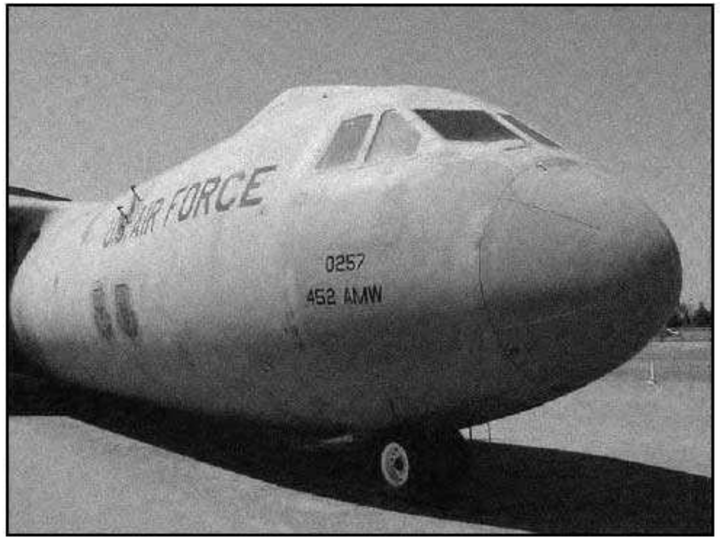
- Improves an image for human interpretation in ways including:
  - Image display and printing
  - Image editting
  - Image enhancement
  - Image compression

Example Operation: Noise Removal

Noisy Image



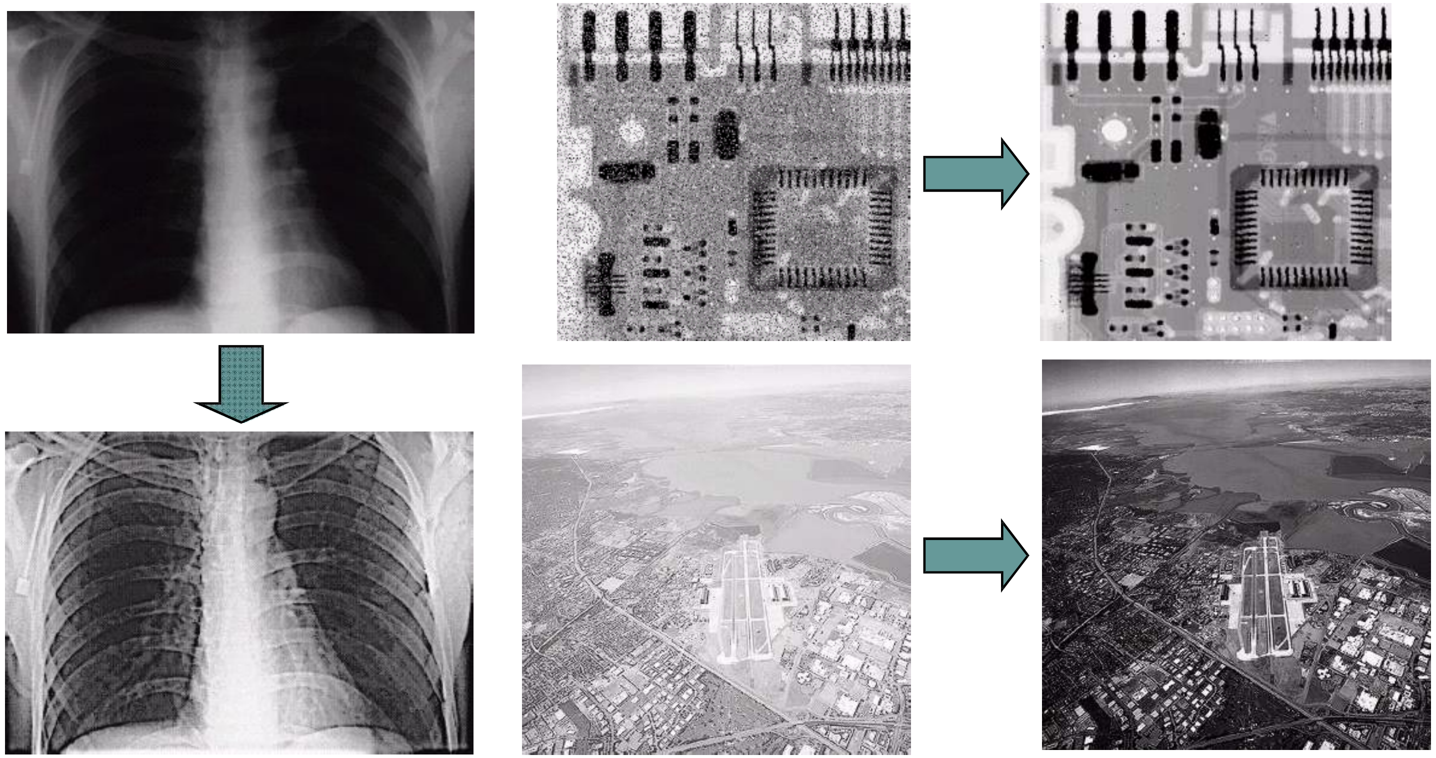
Denoised Image



Think of noise as white specks on a picture (random or non-random)

Examples: Noise Removal

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



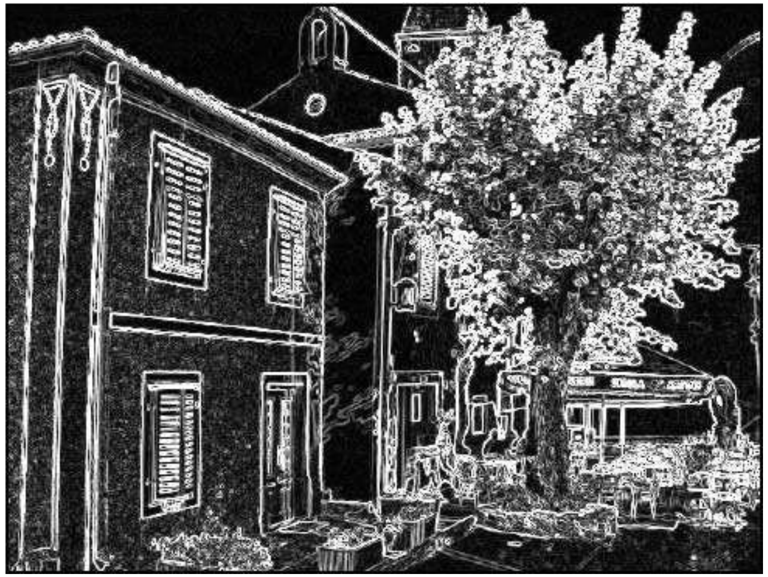
Example: Contrast Adjustment

Low Contrast

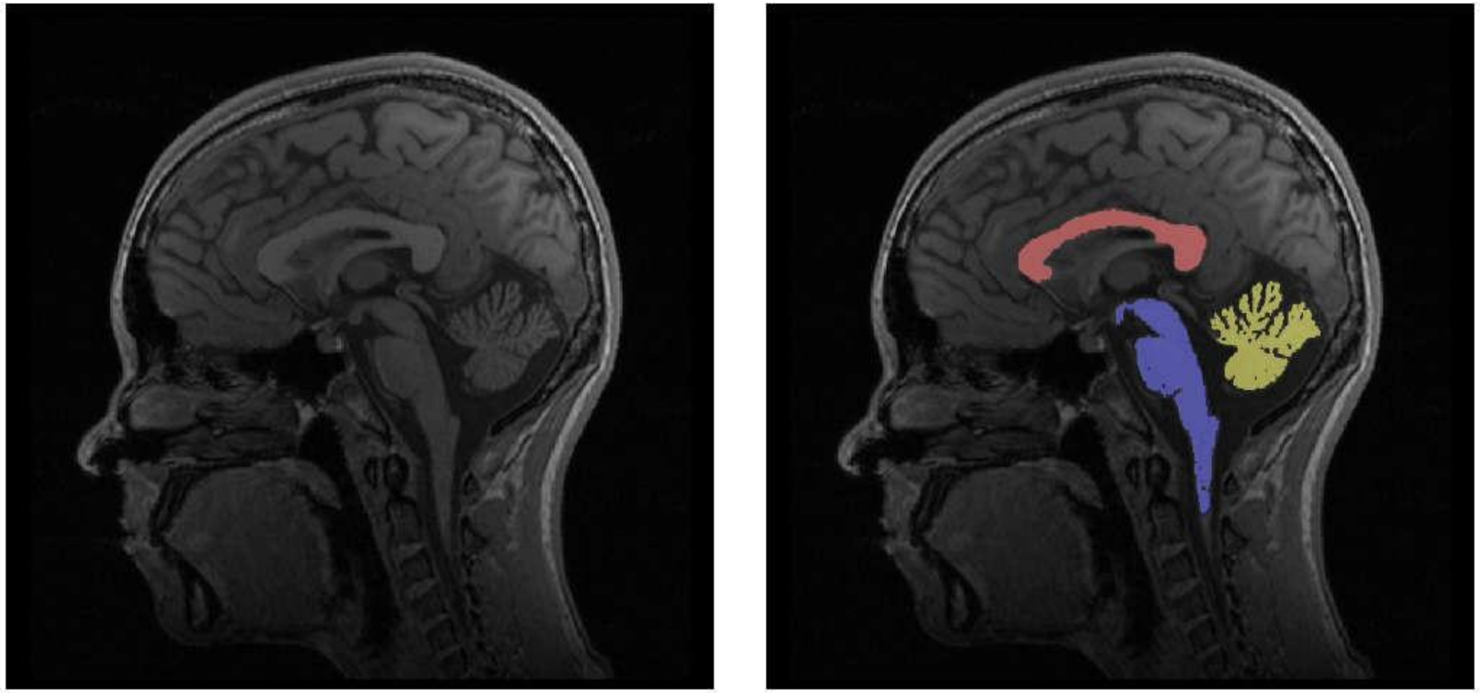
Original Contrast

High Contrast

Example: Edge Detection



Example: Region Detection, Segmentation



Example: Image Compression

Original, 2.1MB

JPEG Compression, 308KB (15%)

Example: Image Inpainting

Damaged Image

Restored Image

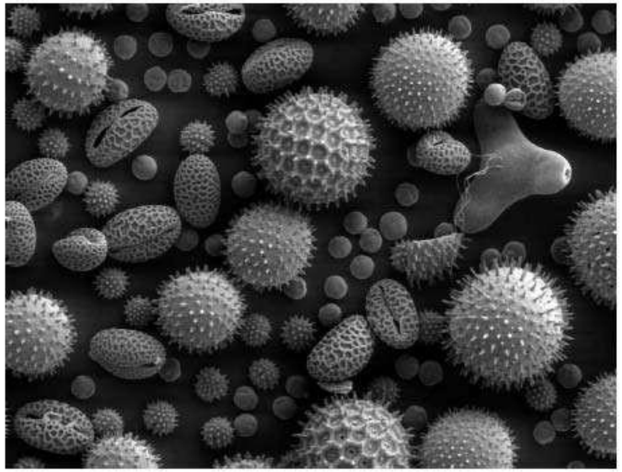
Credit: M. Bertalmio, G. Sapiro, V. Caselles, C. Ballester: Image Inpainting, SIGGRAPH 2000

Inpainting? Reconstruct corrupted/destroyed parts of an image

Examples: Artistic (Movie Special)Effects

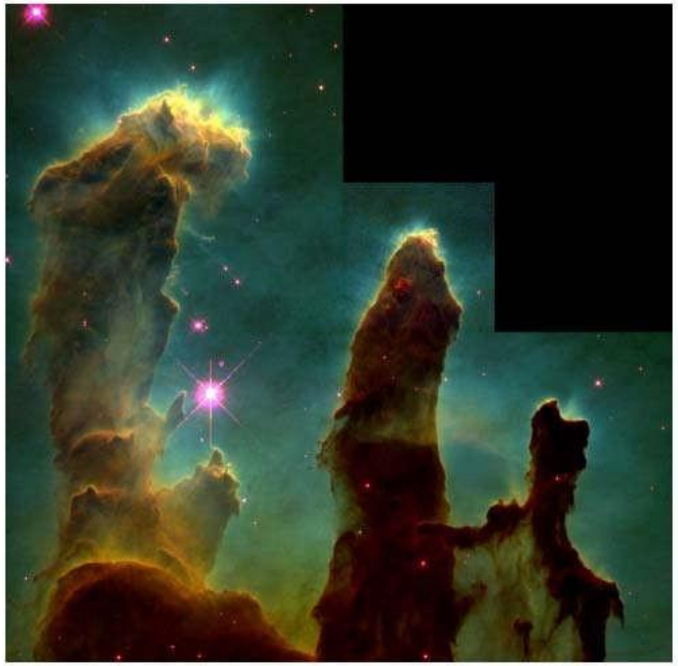
Applications of Image Processing

Biology



Credit: Dartmouth Electron Microscopy Facility

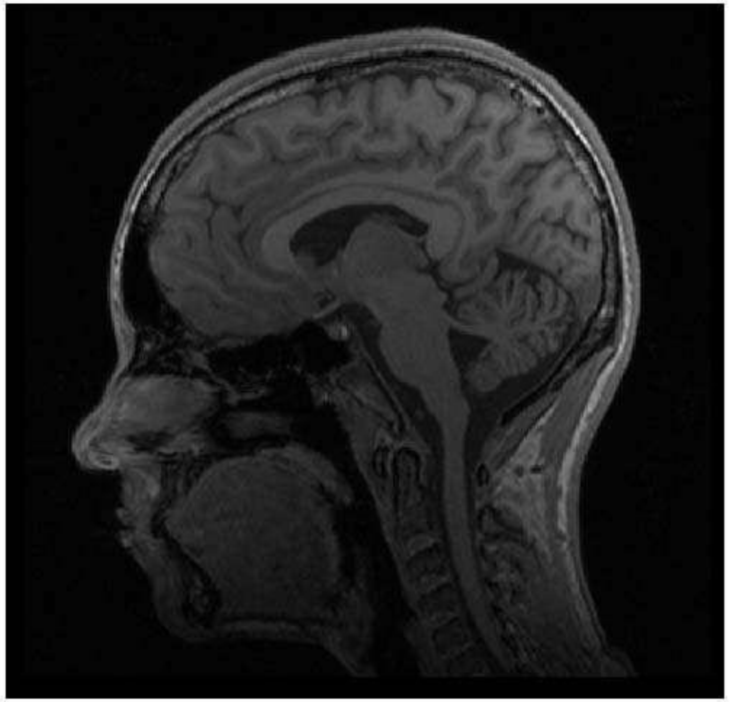
Astronomy



Credit: NASA, Jeff Hester, and Paul Scowen (Arizona State)
More info here

Applications of Image Processing

Medicine

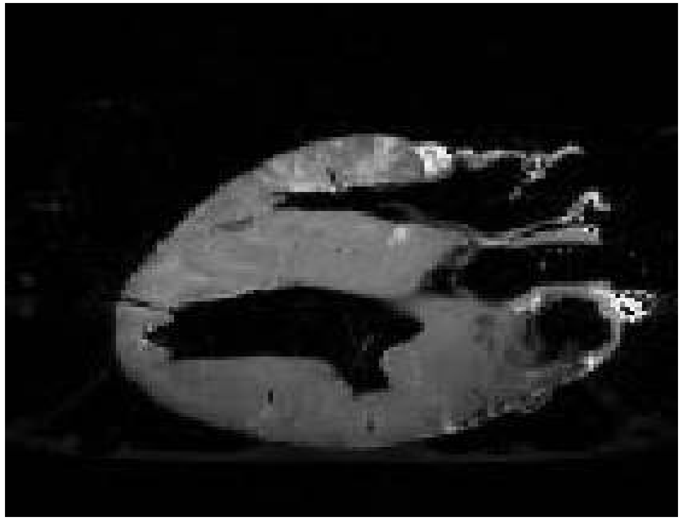


Credit: Dr. Janet Lainhart, UofU Psychiatry

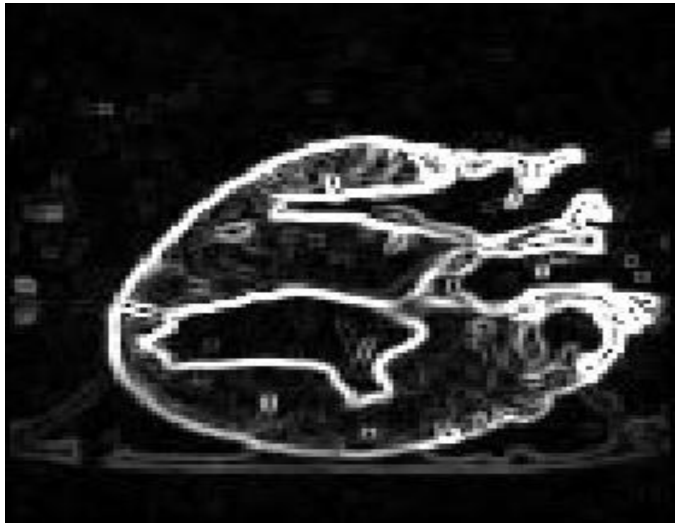
Security, Biometrics

Applications of Image Processing: Medicine

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Original MRI Image of a Dog Heart



Edge Detection Image

Applications of Image Processing

Satellite Imagery

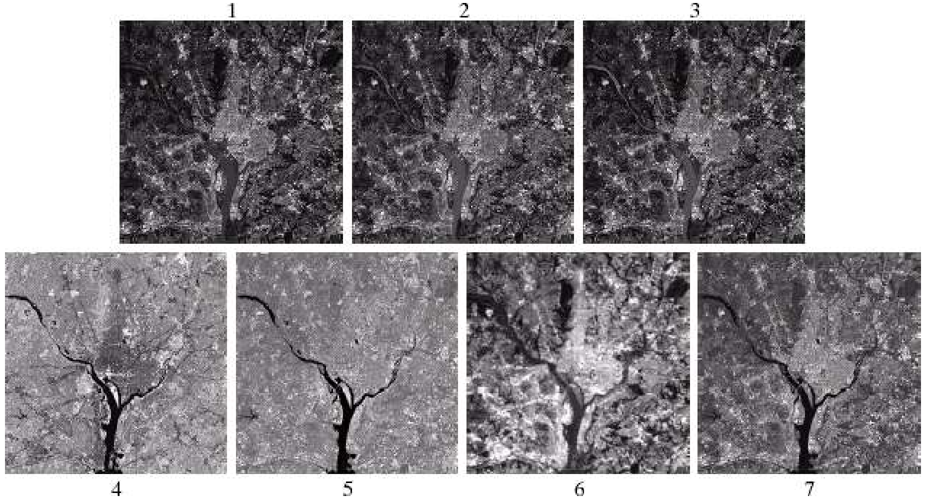
Credit: NASA

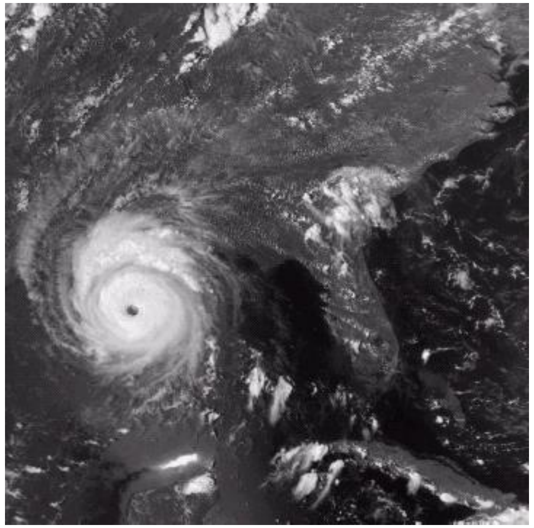
Personal Photos

Credit: Tom Fletcher

Applications of Image Processing: Geographic Information Systems (GIS)

- Terrain classification
- Meteorology (weather)



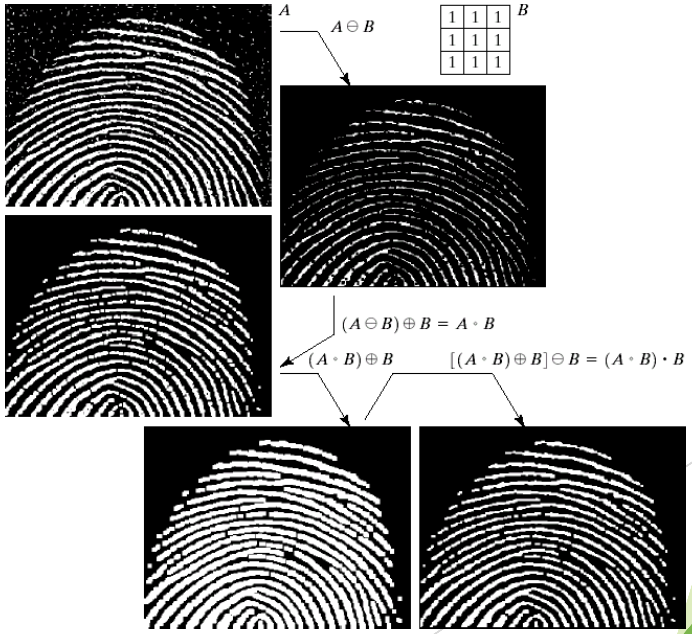


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Applications of Image Processing: Law Enforcement

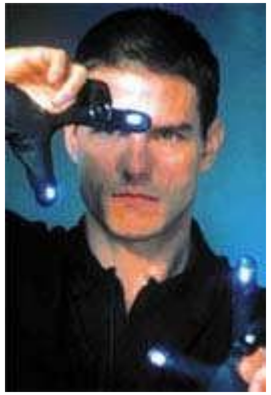
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

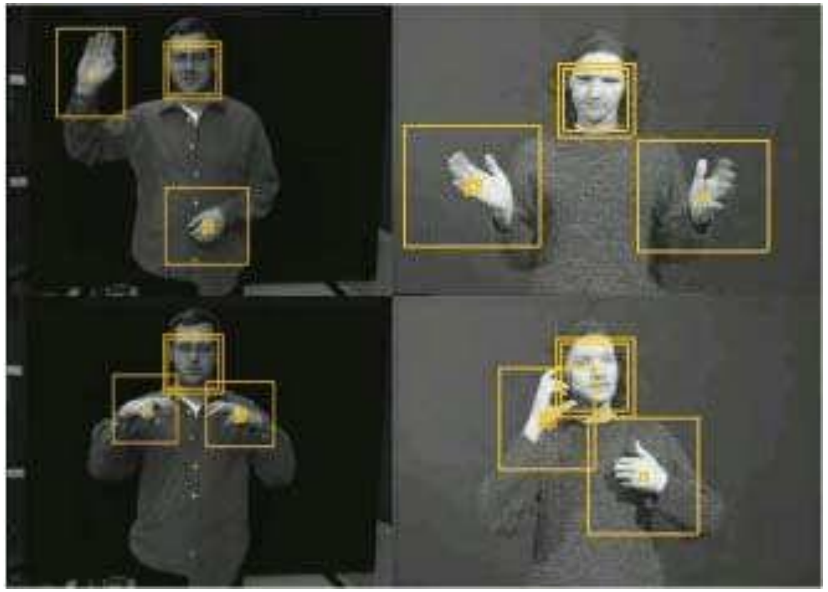
- Number plate recognition for speed cameras or automated toll systems
- Fingerprint recognition

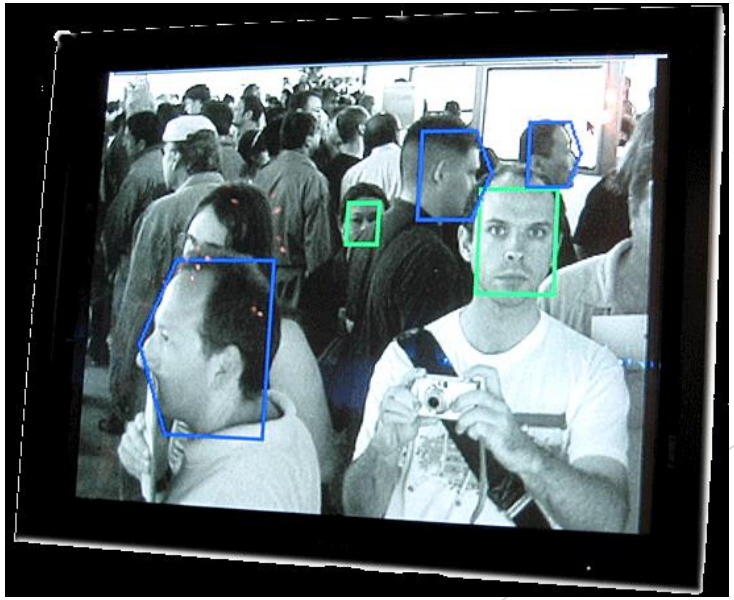


Applications of Image Processing: HCI

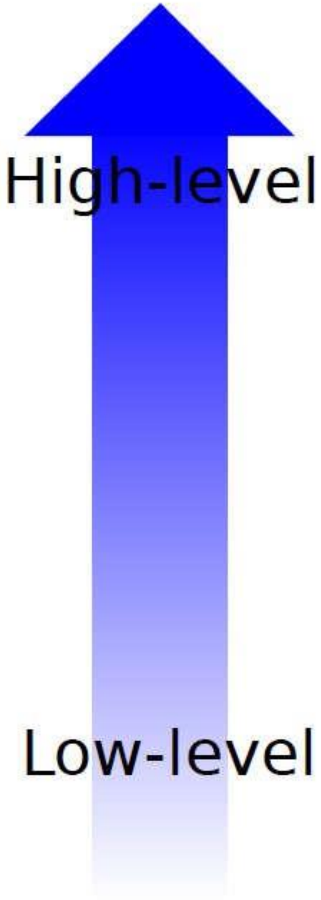
- Face recognition
- Gesture recognition







Relationship with other Fields



Computer Vision

Object detection, recognition, shape analysis, tracking
Use of Artificial Intelligence and Machine Learning

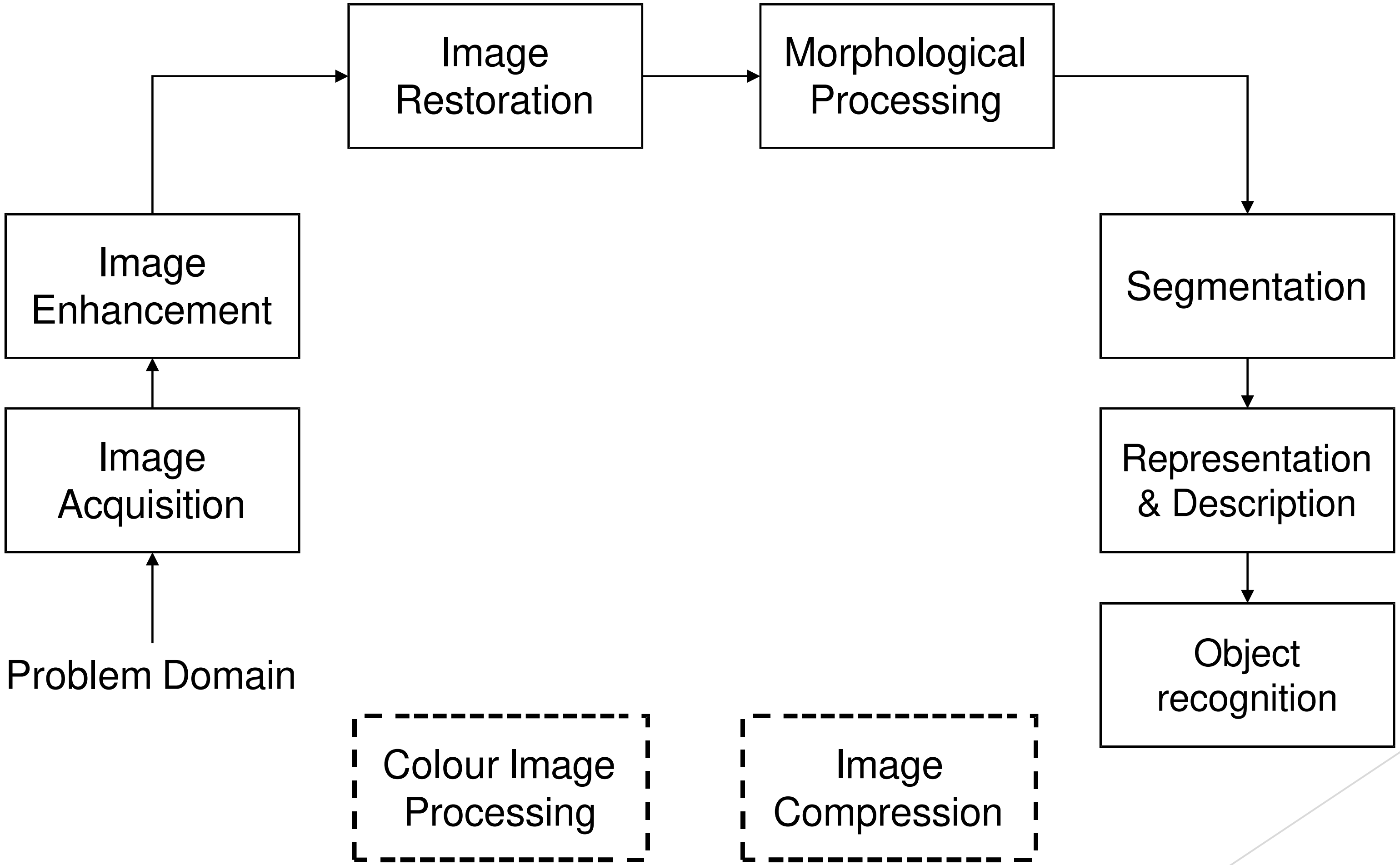
Image Analysis

Segmentation, image registration, matching

Image Processing

Image enhancement, noise removal, restoration,
feature detection, compression

Key Stages in Digital Image Processing



Key Stages in Digital Image Processing: Image Aquisition

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Restoration

Morphological Processing

Image Enhancement

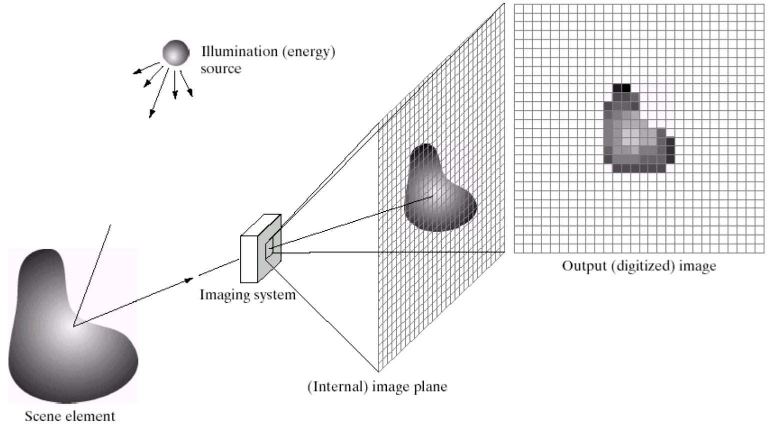
Segmentation

Image Acquisition

Representation & Description

Problem Domain

Object recognition



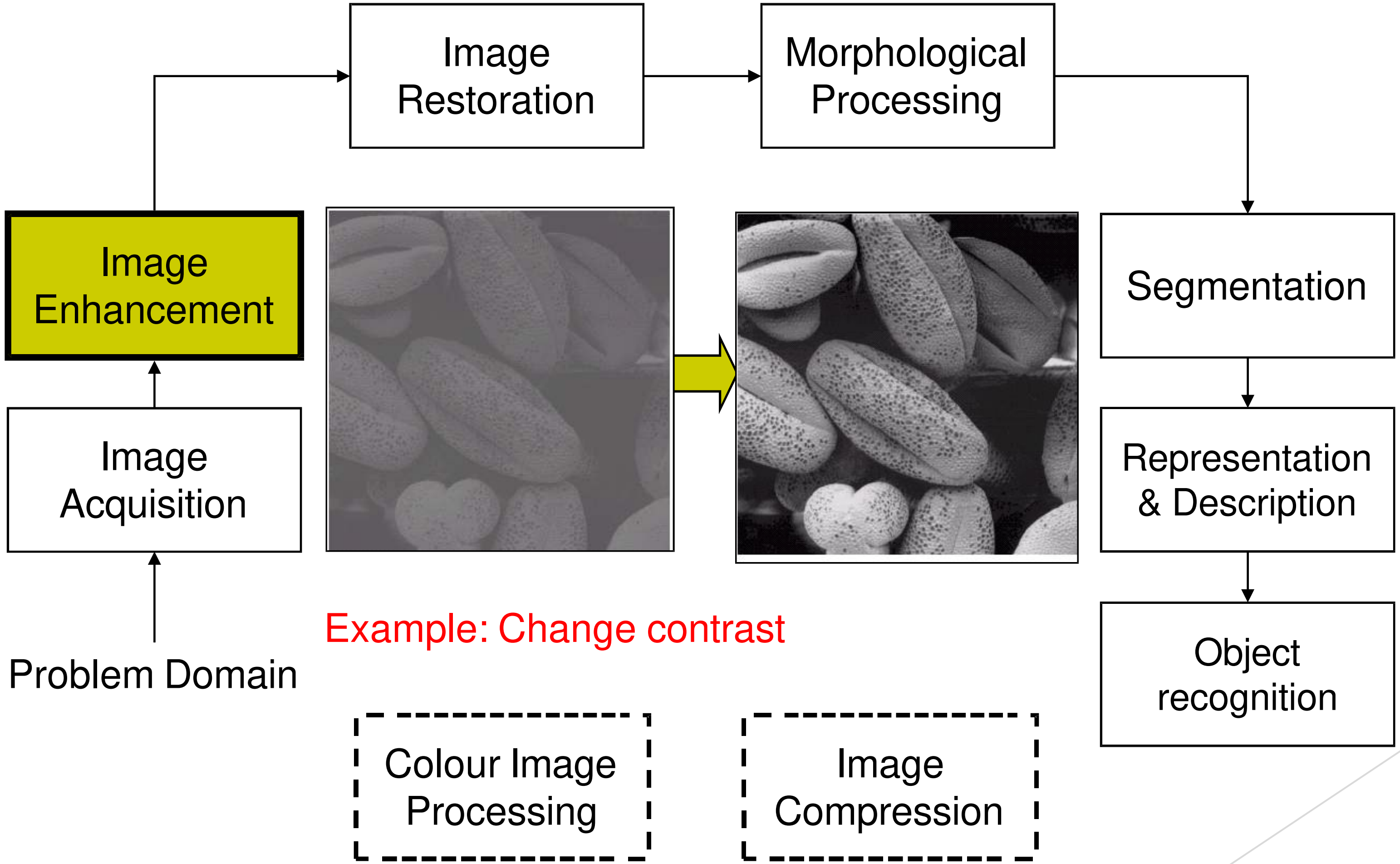
Example: Take a picture

Colour Image Processing

Image Compression

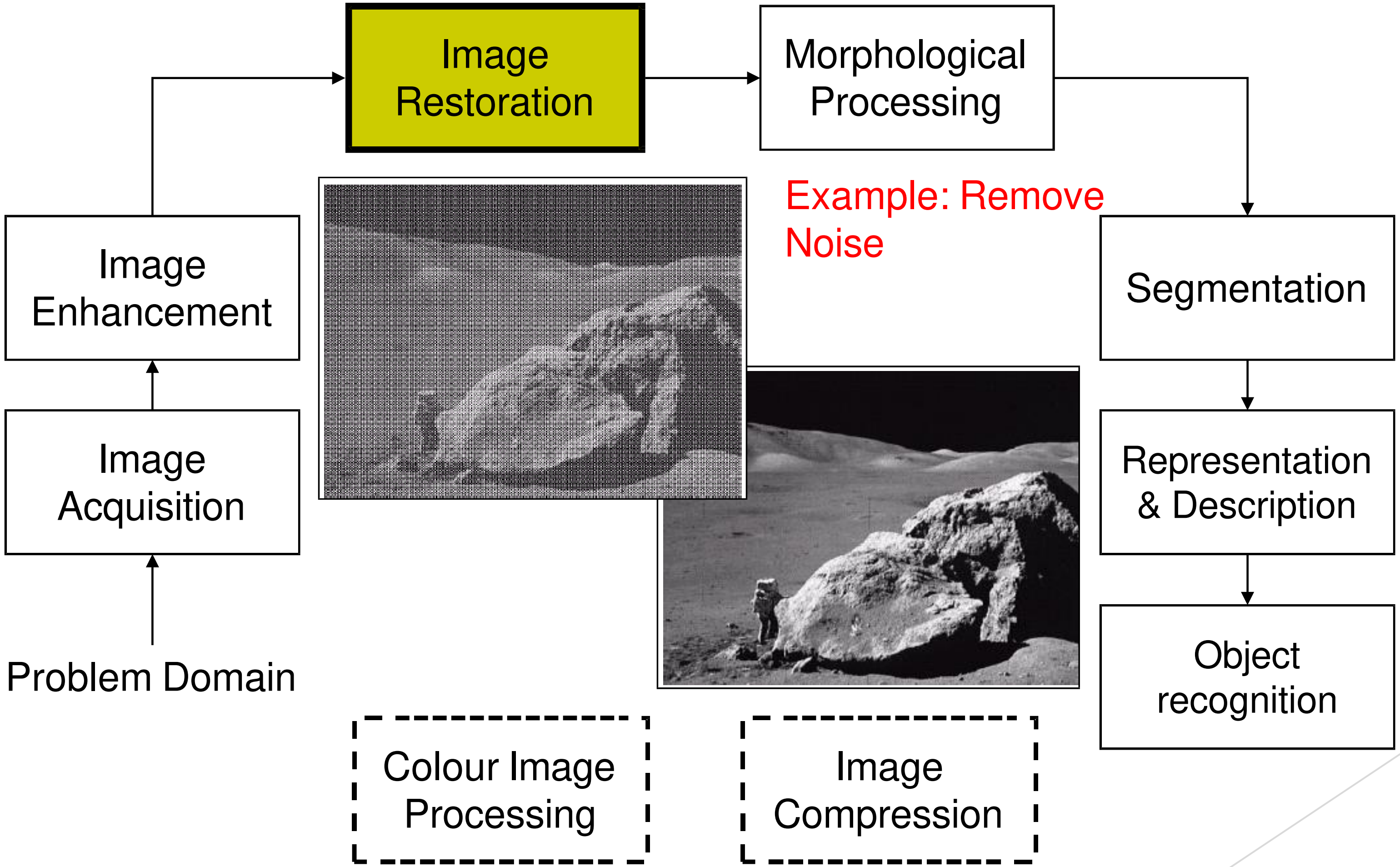
Key Stages in Digital Image Processing: Image Enhancement

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



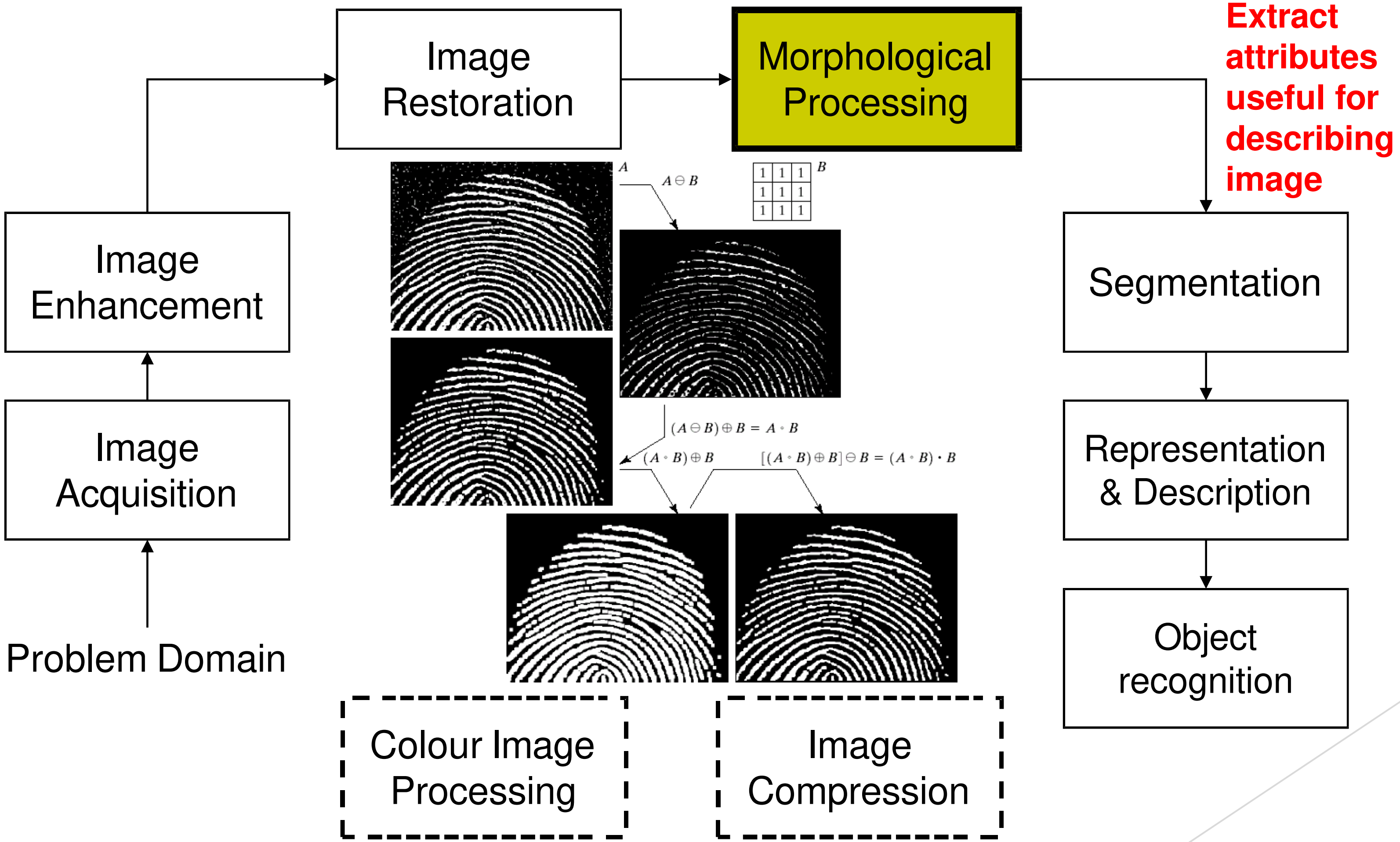
Key Stages in Digital Image Processing: Image Restoration

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



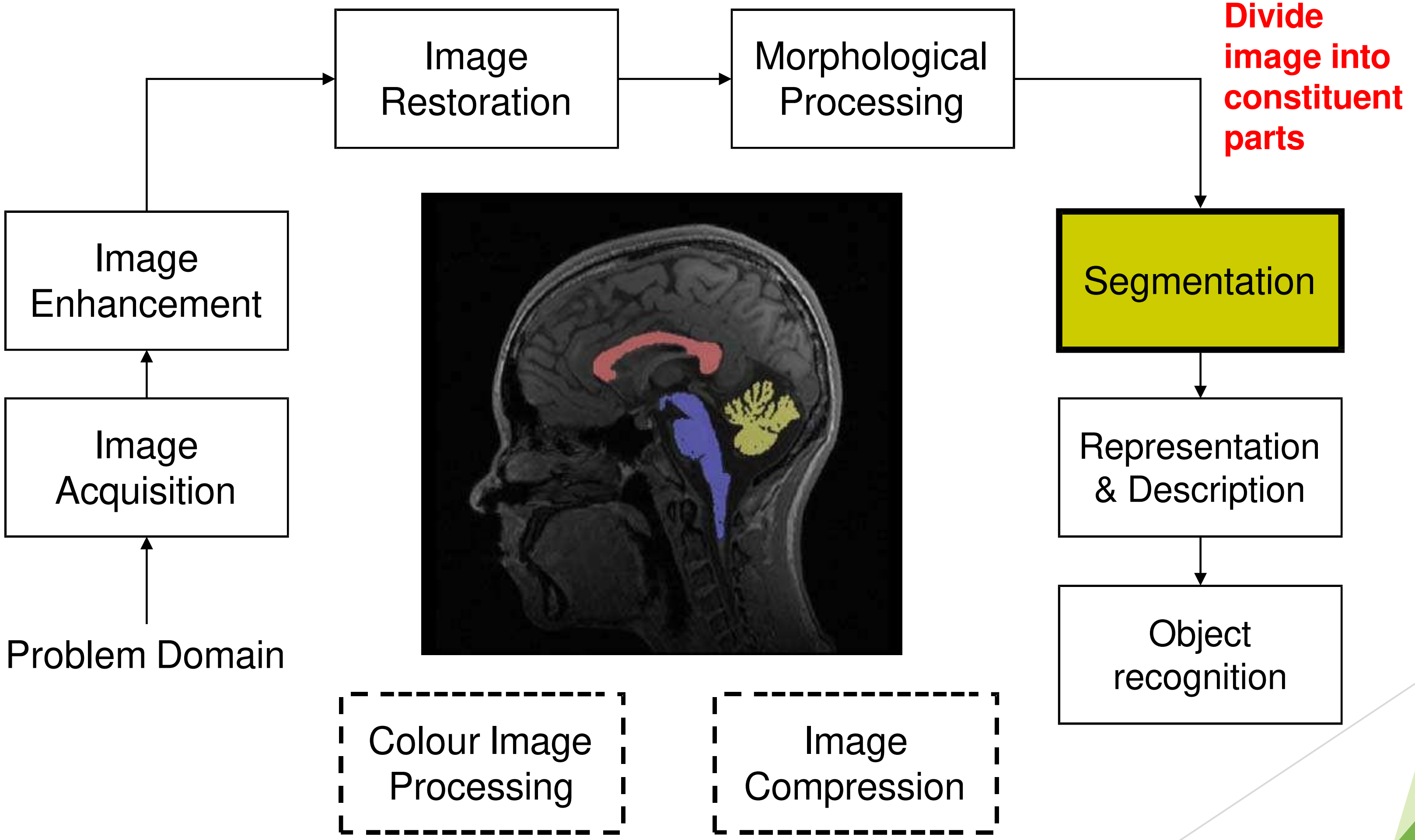
Key Stages in Digital Image Processing: Morphological Processing

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Key Stages in Digital Image Processing: Segmentation

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Key Stages in Digital Image Processing: Object Recognition

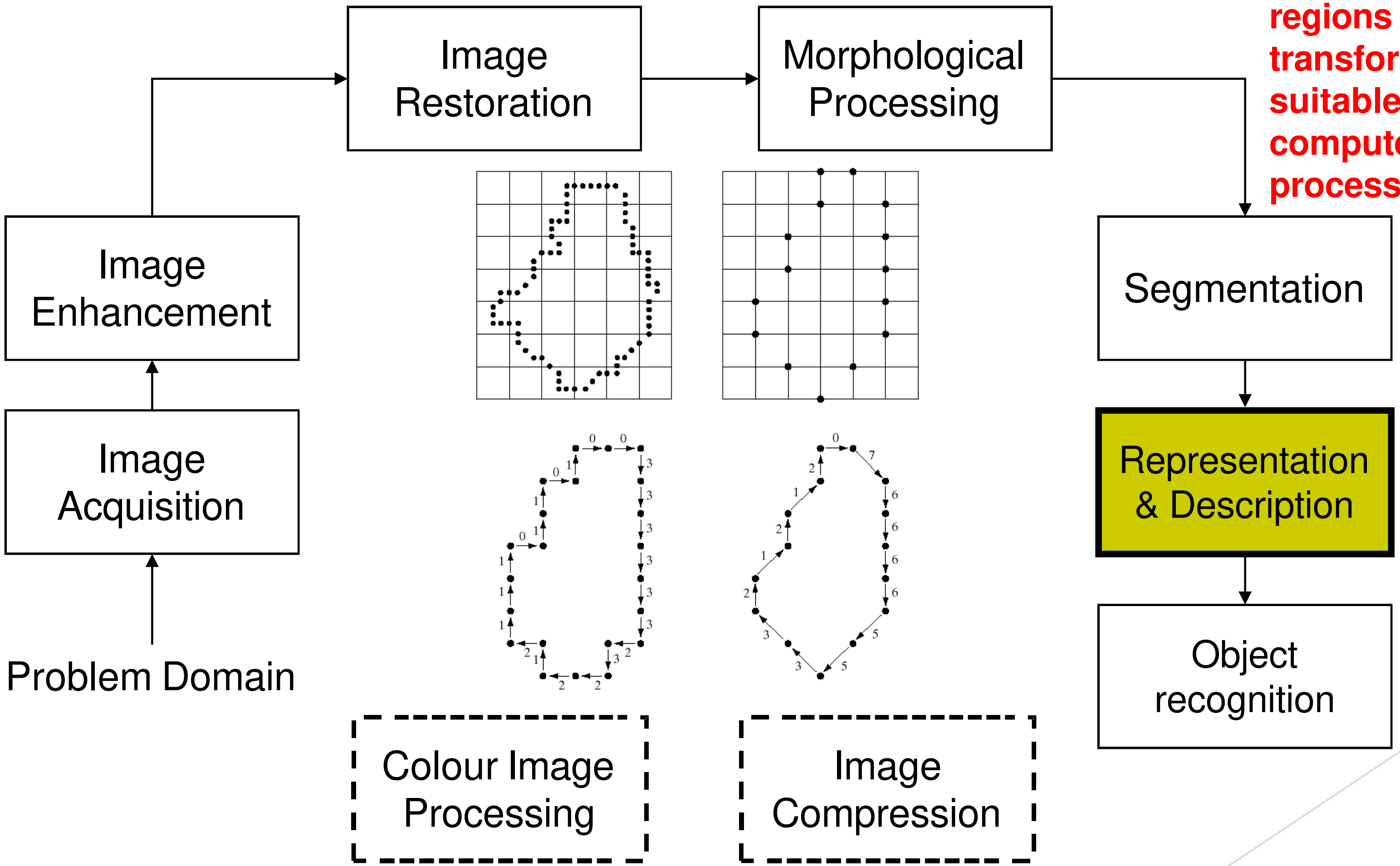
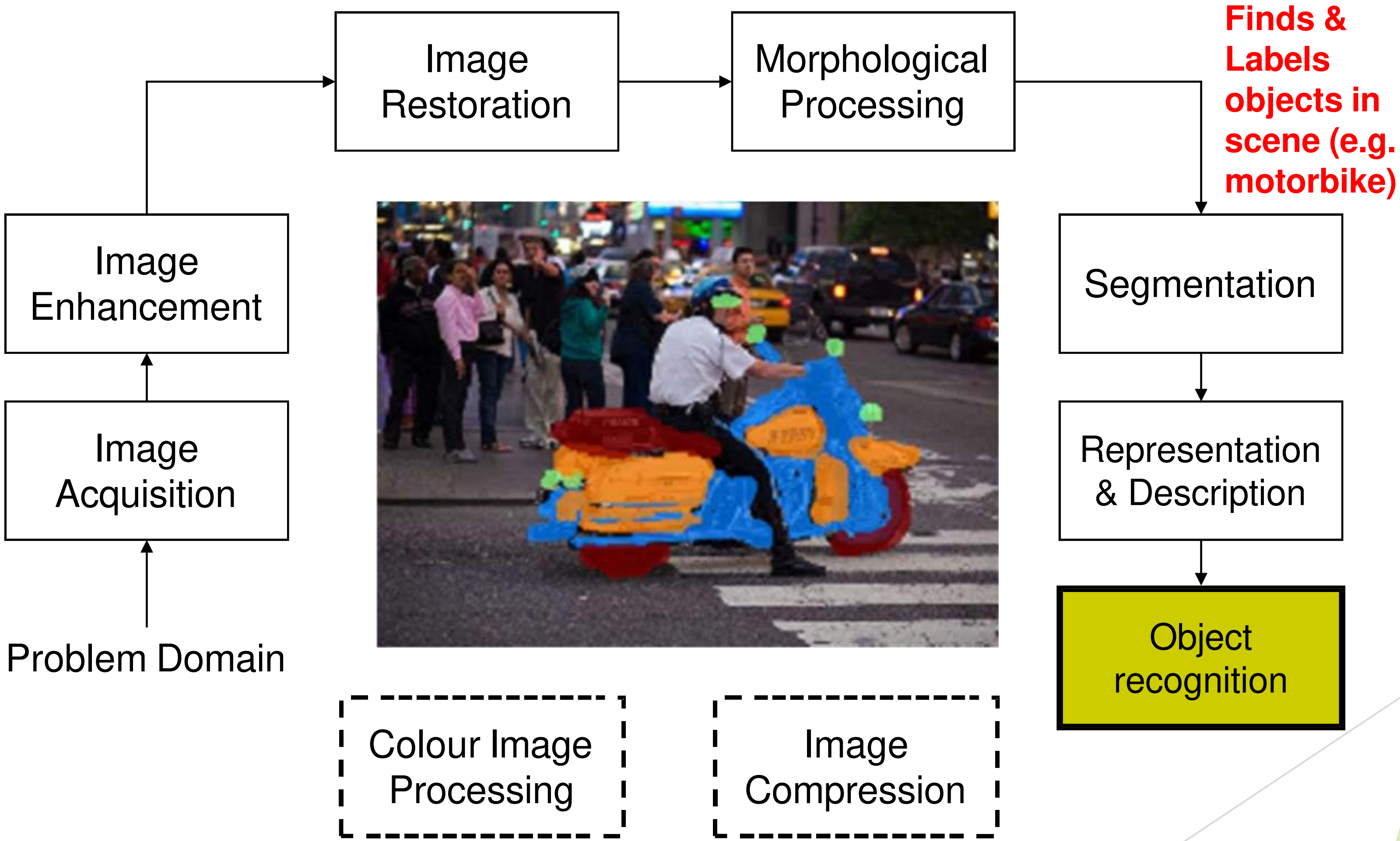


Image regions transformed suitable for computer processing

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Key Stages in Digital Image Processing: Representation & Description



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Key Stages in Digital Image Processing: Image Compression

Image Restoration

Morphological Processing

Reduce image size (e.g. JPEG)

Image Enhancement

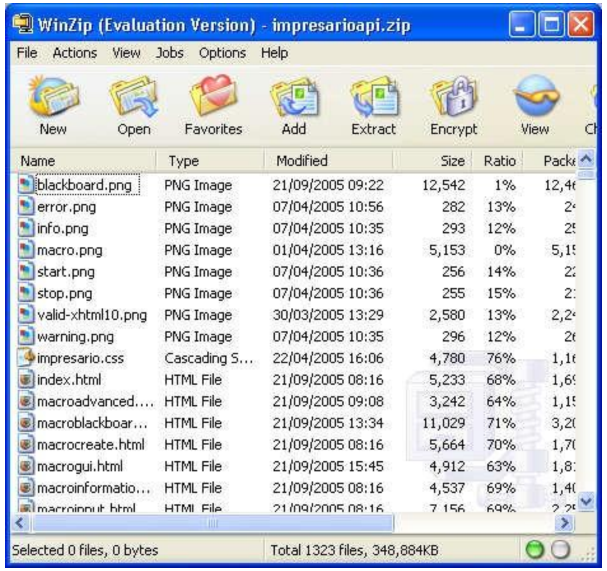
Segmentation

Image Acquisition

Representation & Description

Problem Domain

Object recognition



Colour Image Processing

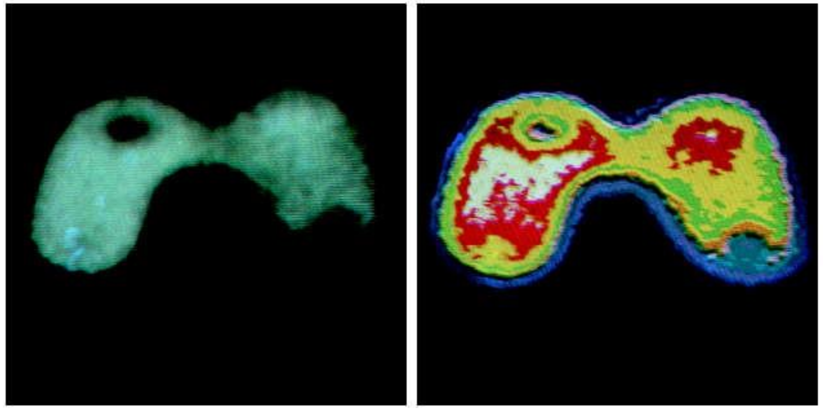
Image Compression

Key Stages in Digital Image Processing: Colour Image Processing

Image Restoration

Morphological Processing

Image Enhancement



Segmentation

Image Acquisition

Representation & Description

Object recognition

Problem Domain

Consider color images (color models, etc)

Colour Image Processing

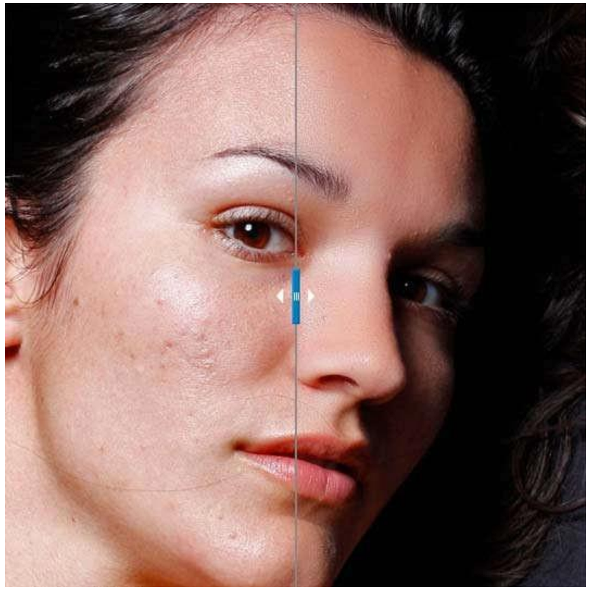
Image Compression

Mathematics for Image Processing

- Calculus
- Linear algebra
- Probability and statistics
- Differential Equations (PDEs and ODEs)
- Differential Geometry
- Harmonic Analysis (Fourier, wavelet, etc)

About This Course

- Image Processing has many aspects
  - Computer Scientists/Engineers develop tools (e.g. photoshop)
    - Requires knowledge of maths, algorithms, programming
  - Artists use image processing tools to modify pictures
    - DOES NOT require knowledge of maths, algorithms, programming



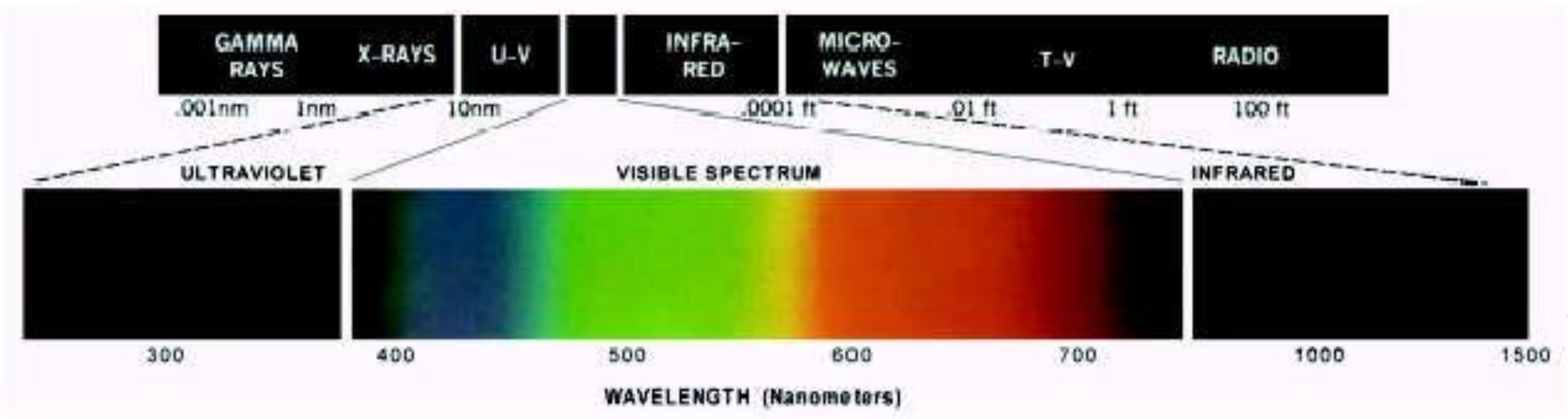
Example: Portraiture photoshop plugin

Example: Knoll Light Factory photoshop plugin

Example: ToonIt photoshop plugin

Light And The Electromagnetic Spectrum

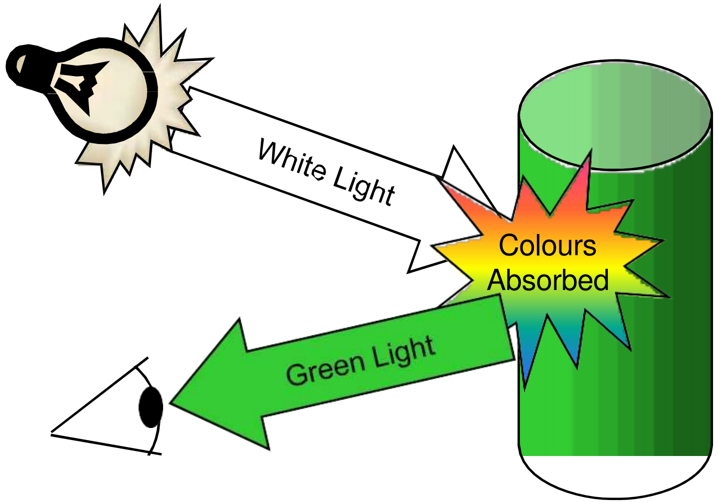
- Light: just a particular part of electromagnetic spectrum that can be sensed by the human eye
- The electromagnetic spectrum is split up according to the wavelengths of different forms of energy



Reflected Light

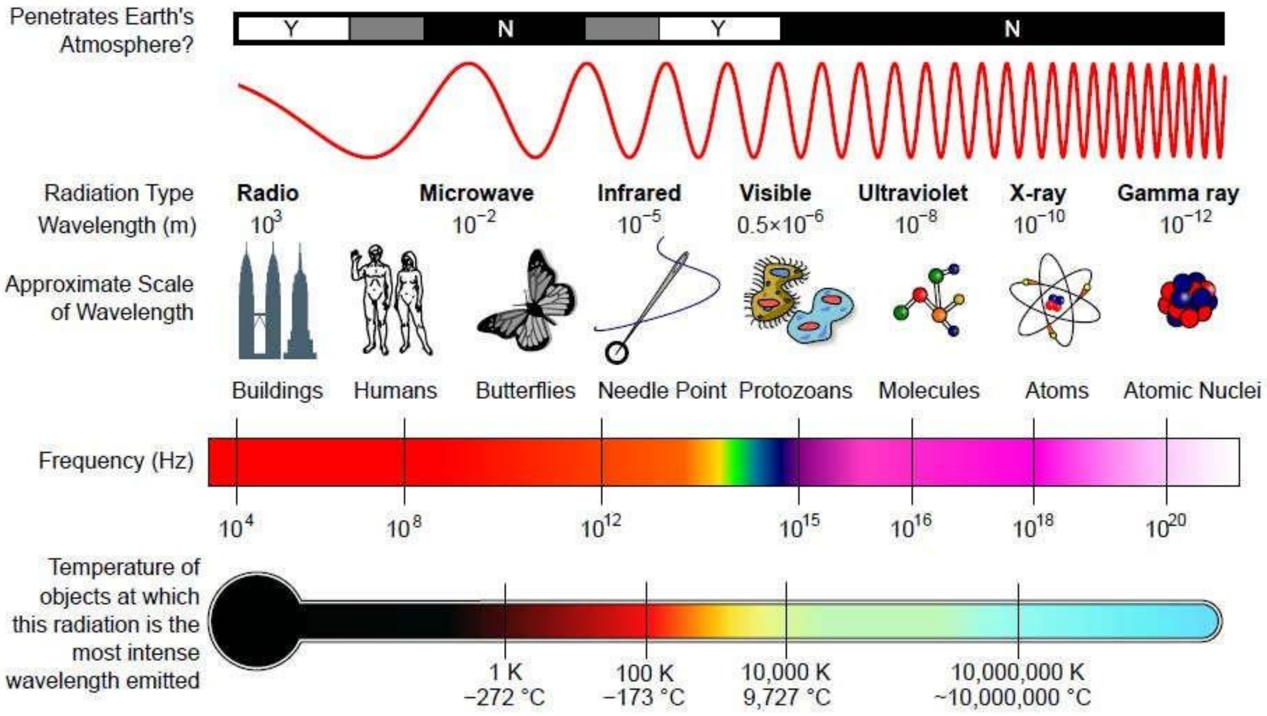
- The colours humans perceive are determined by nature of light reflected from an object

- For example, if white light (contains all wavelengths) is shone onto green object it absorbs most wavelengths absorbed except green wavelength (color)



Electromagnetic Spectrum and IP

- Images can be made from any form of EM radiation



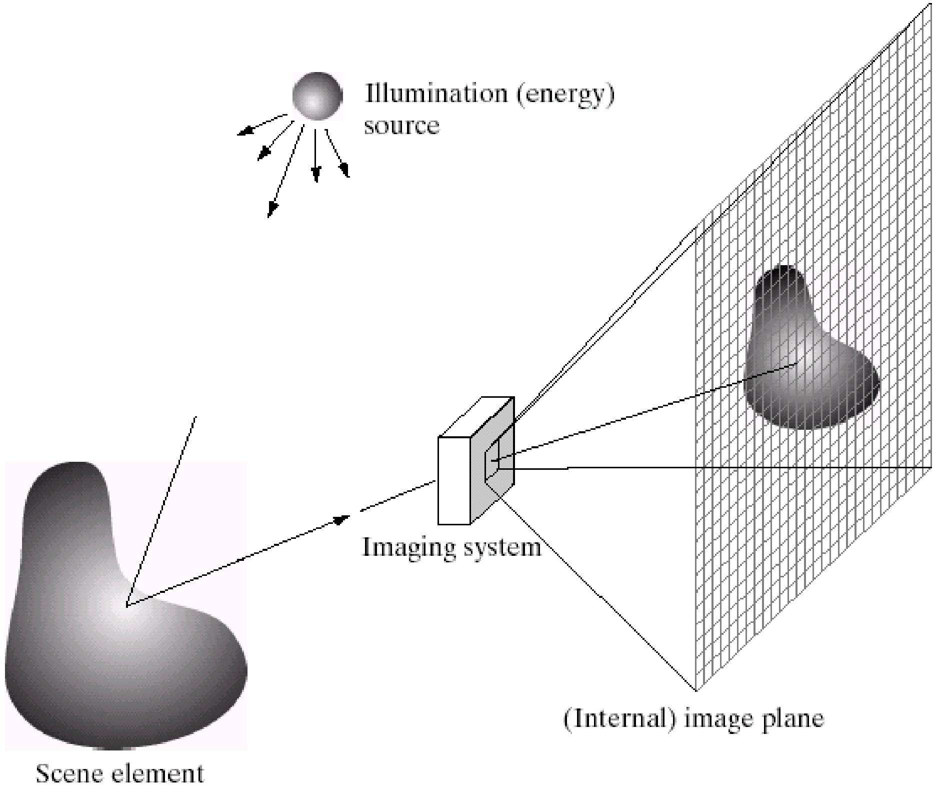
From Wikipedia

Images from Different EM Radiation

- Radar imaging (radio waves)
- Magnetic Resonance Imaging (MRI) (Radio waves)
- Microwave imaging
- Infrared imaging
- Photographs
- Ultraviolet imaging telescopes
- X-rays and Computed tomography
- Positron emission tomography (gamma rays)
- Ultrasound (not EM waves)

Image Acquisition

- Images typically generated by *illuminating a scene* and absorbing energy reflected by scene objects



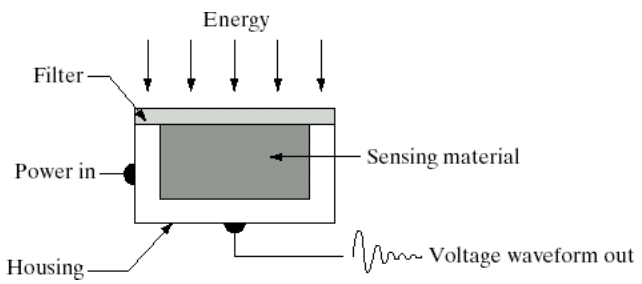
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Sensing

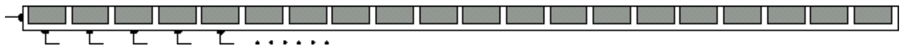
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

- Incoming energy (e.g. light) lands on a sensor material responsive to that type of energy, generating a voltage

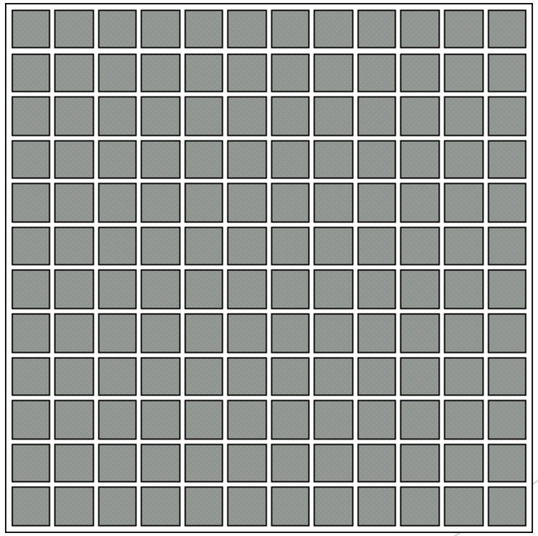
- Collections of sensors are arranged to capture images



Imaging Sensor



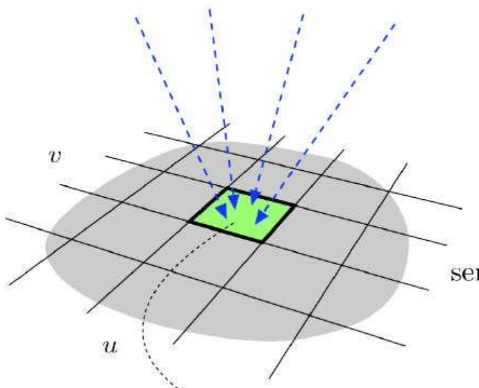
Line of Image Sensors



Array of Image Sensors

Spatial Sampling

- Cannot record image values for all (x,y)
- Sample/record image values at discrete (x,y)
- Sensors arranged in grid to sample image



$F(x, y)$

148	123	52	107	123	162	172	123	64	89	⋯
147	130	92	95	98	130	171	155	169	163	⋯
141	118	121	148	117	107	144	137	136	134	⋯
82	106	93	172	149	131	138	114	113	129	⋯
57	101	72	54	109	111	104	135	106	125	⋯
138	135	114	82	121	110	34	76	101	111	⋯
138	102	128	159	168	147	116	129	124	117	⋯
113	89	89	109	106	126	114	150	164	145	⋯
120	121	123	87	85	70	119	64	79	127	⋯
145	141	143	134	111	124	117	113	64	112	⋯
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

$I(u, v)$

Image (Spatial) Sampling

- A digital sensor can only measure a limited number of samples at a discrete set of energy levels
- Sampling can be thought of as:

Continuous signal x comb function

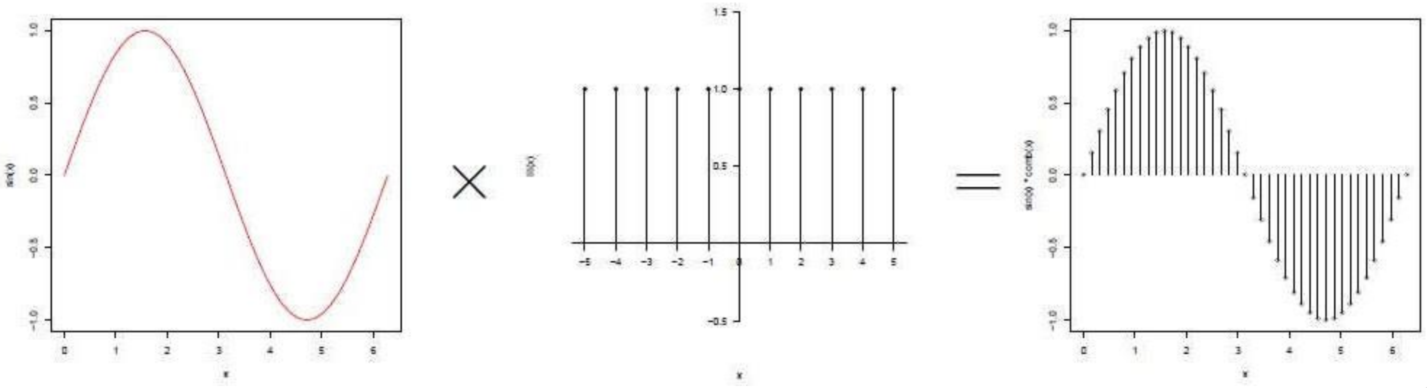


Image Quantization

- *Quantization:* process of converting continuous analog signal into its digital representation
- Discretize image *I(u,v)* values
- Limit values image can take

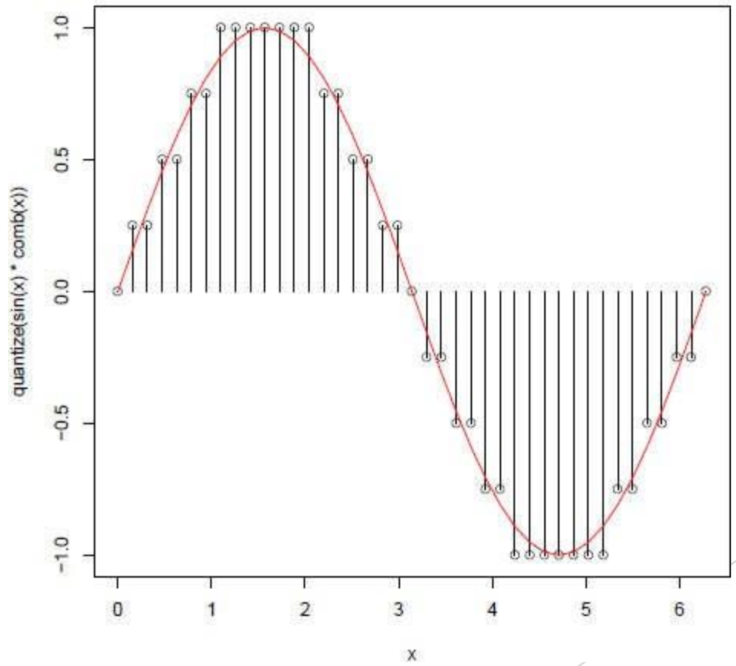


Image Sampling And Quantization

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

- Sampling and quantization generates approximation of a real world scene

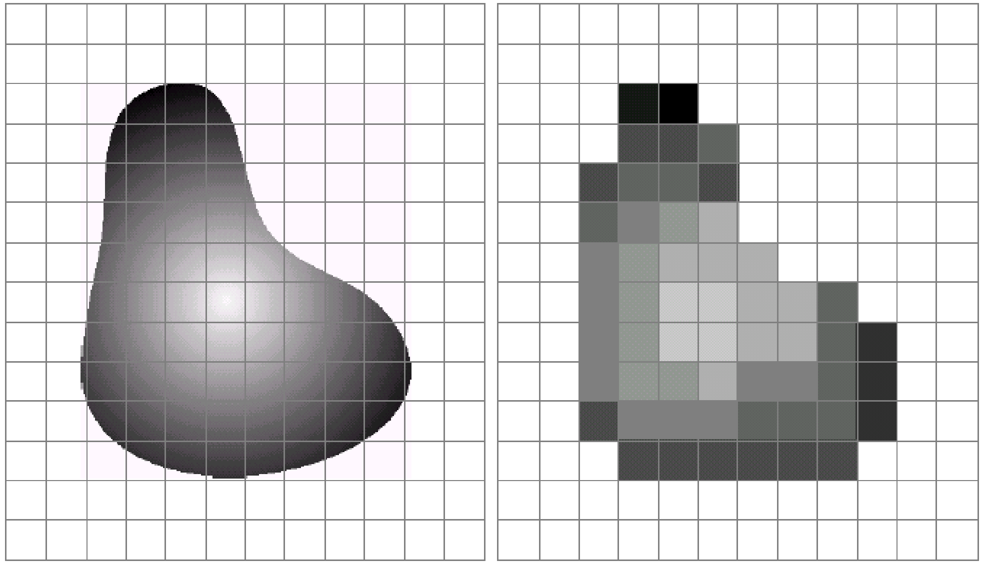


Image as Discrete Function

After spatial sampling and quantization, an image is a discrete function. The image domain Ω is now discrete:

$$\Omega \subset \mathbb{N}^2,$$

and so is the image range:

$$I : \Omega \to \{1, \ldots, K\},$$

where $K \in \mathbb{N}$.

Image as a Function

A simple image

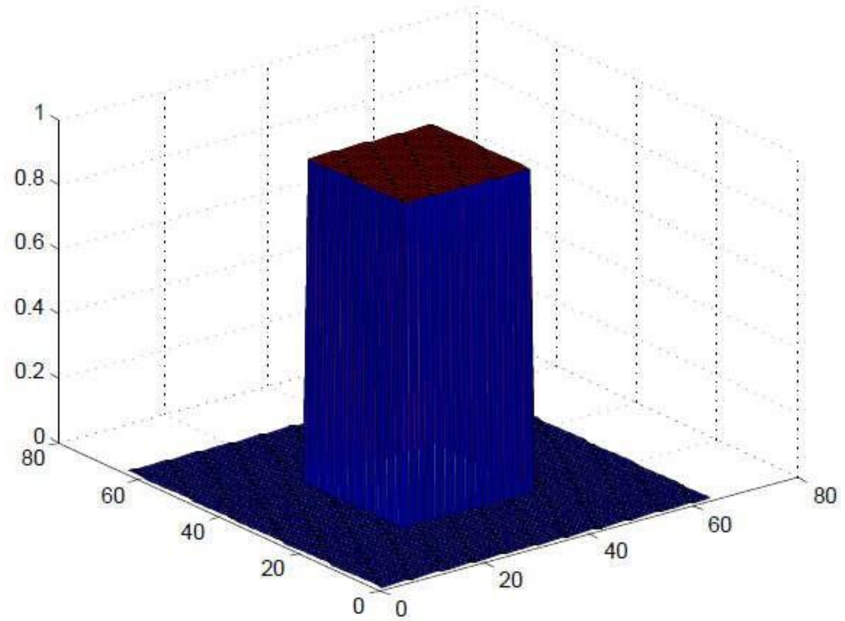
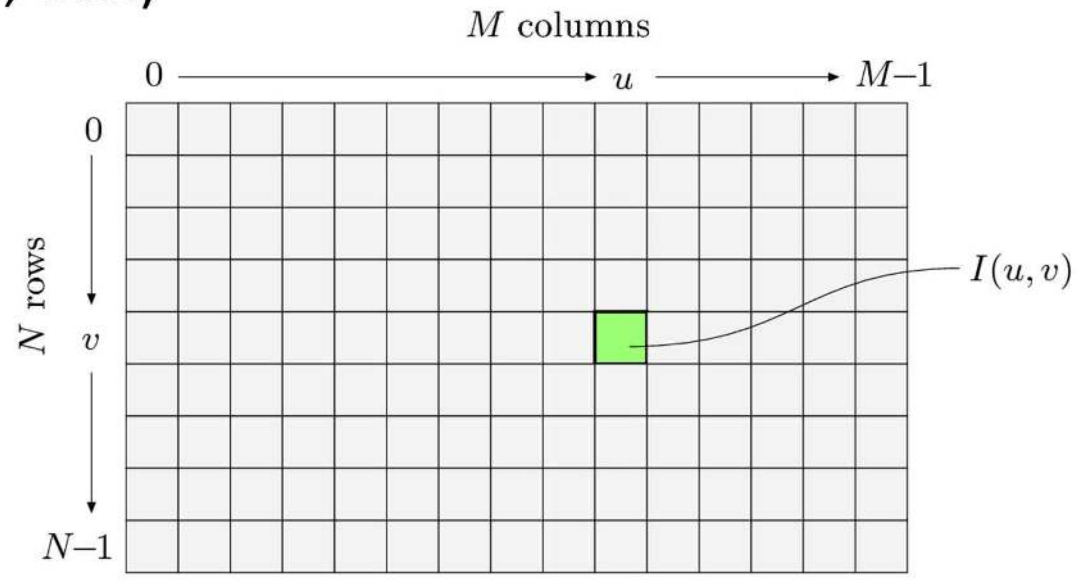


Image function as a height field

Representing Images

- Image data structure is 2D array of pixel values
- Pixel values are gray levels in range 0-255 or RGB colors
- Array values can be any data type (bit, byte, int, float, double, etc.)

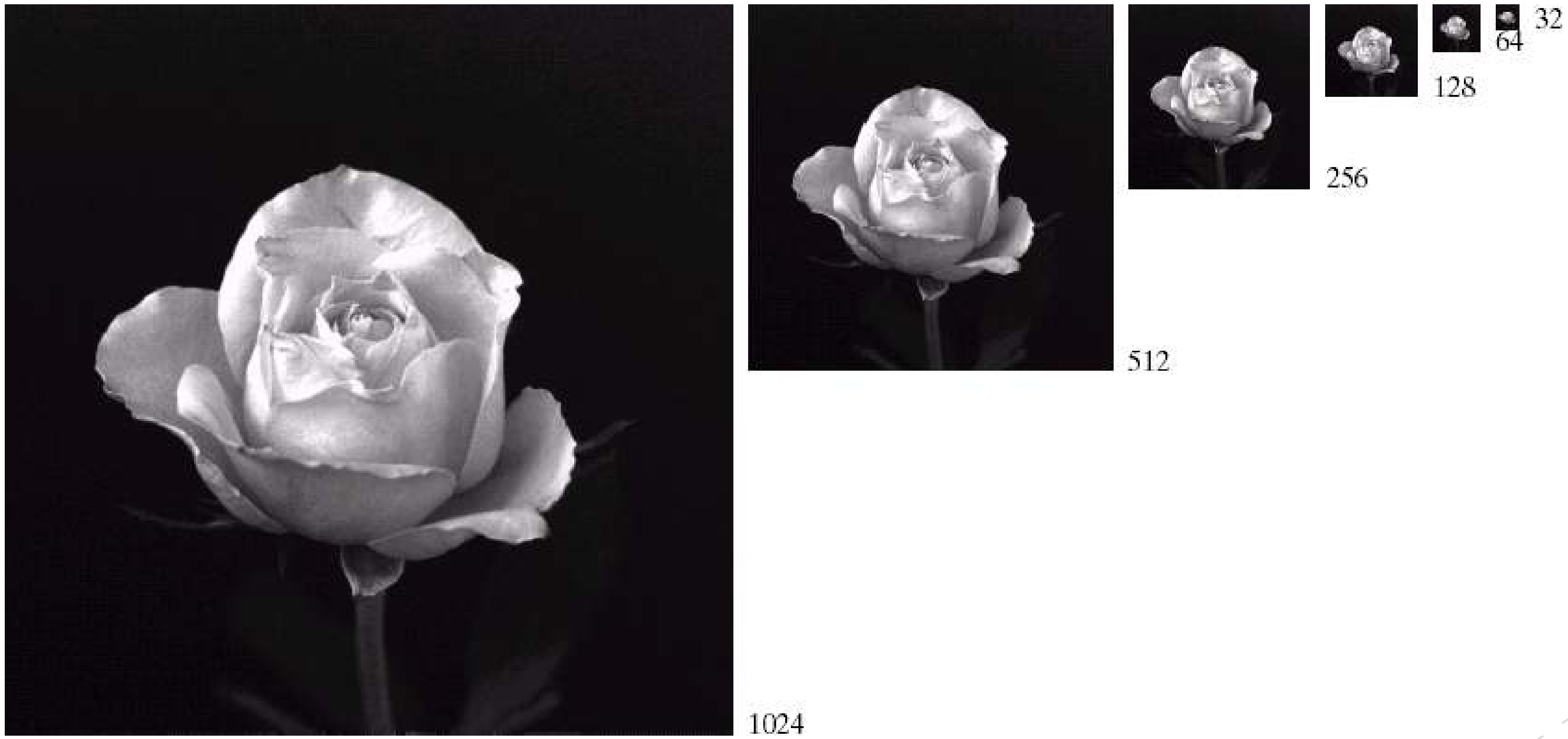


Spatial Resolution

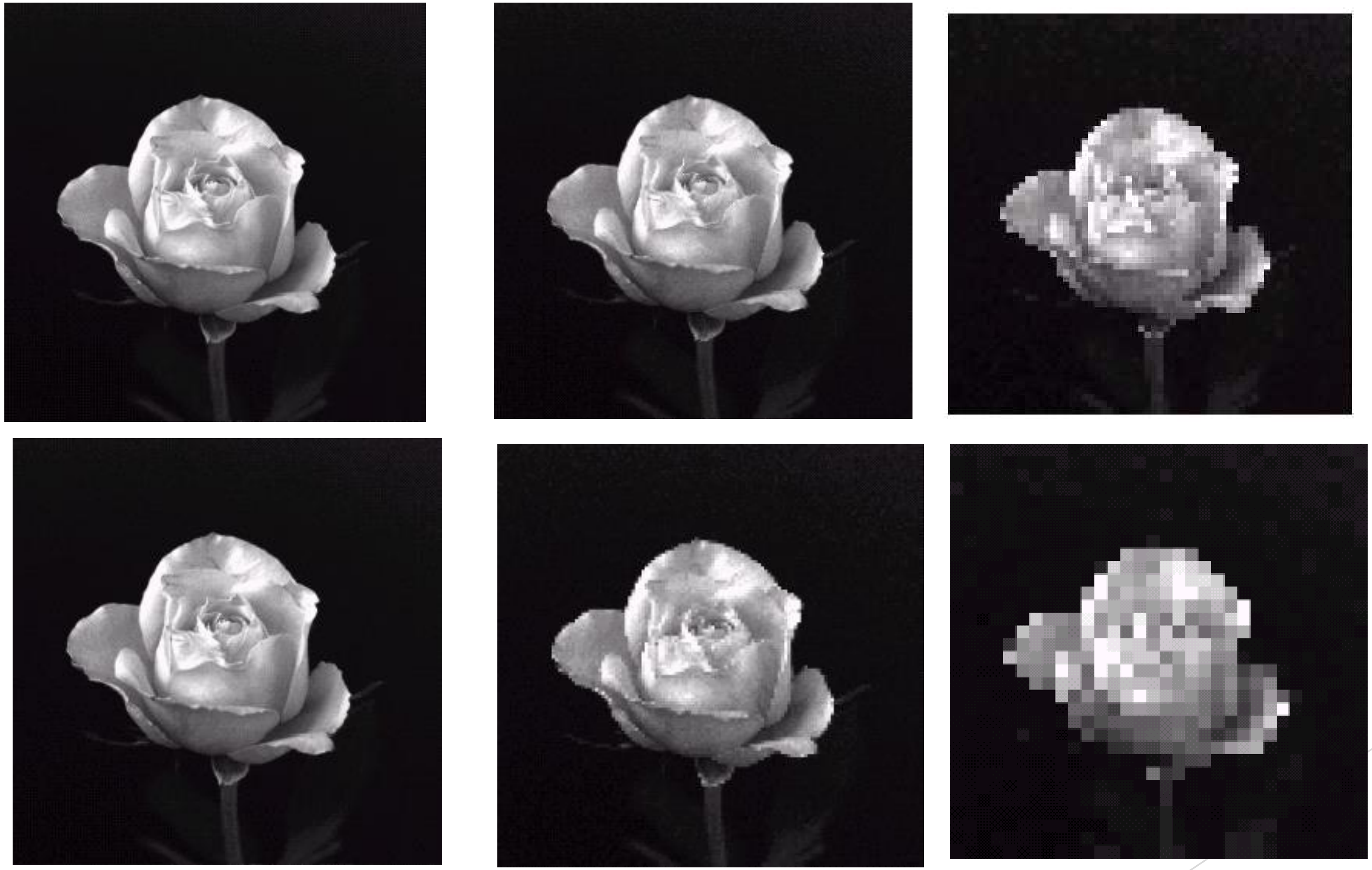
- *The* *spatial resolution* of an image is determined by how fine/coarse sampling was carried out
- Spatial resolution: smallest discernable image detail
  - Vision specialists
  - talk about image resolution
  - Graphic designers talk about *dots per inch* (DPI)

Spatial Resolution

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Spatial Resolution: Stretched Images



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

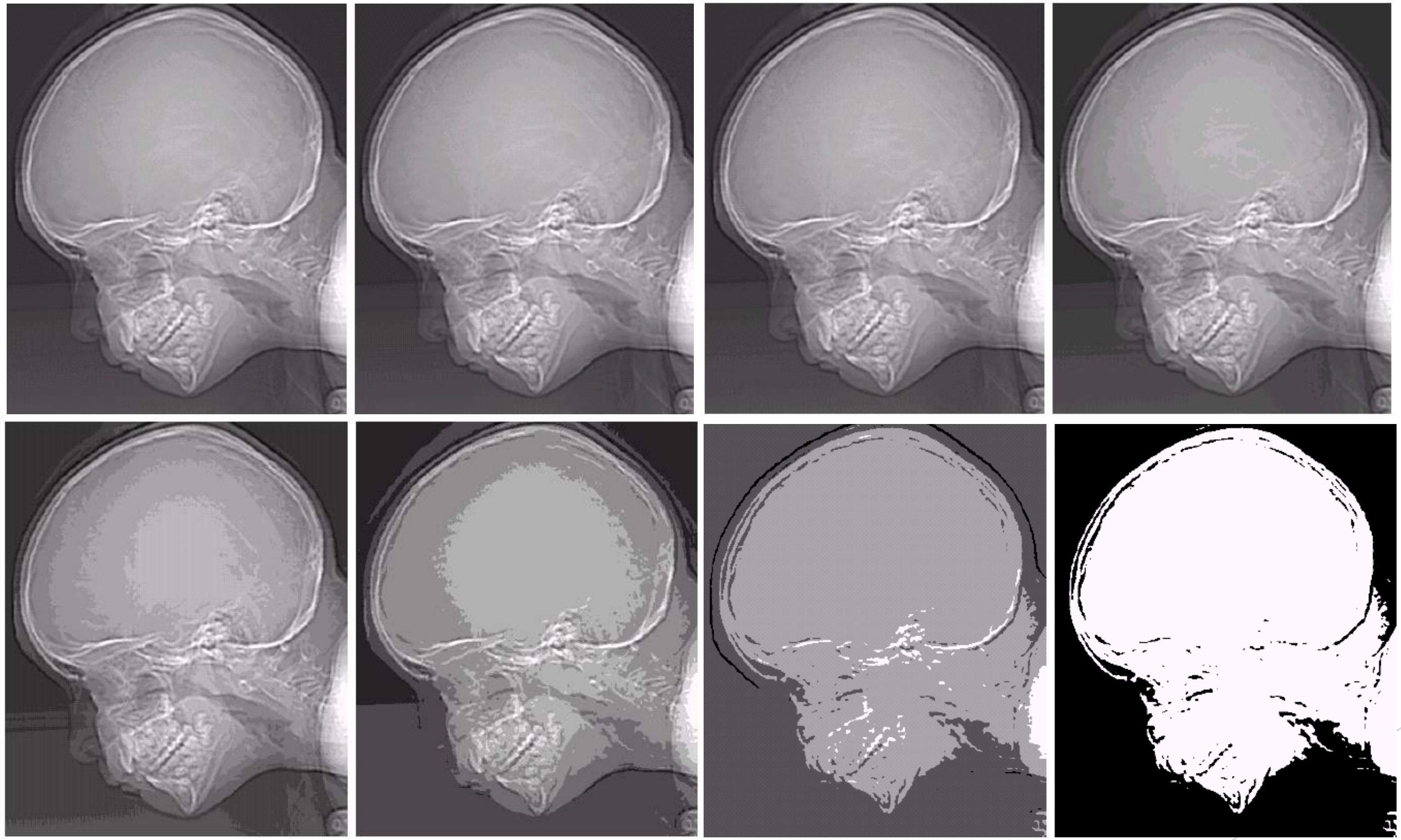
Intensity Level Resolution

- *Intensity level resolution*: number of intensity levels used to represent the image
  - The more intensity levels used, the finer the level of detail discernable in an image
  - Intensity level resolution usually given in terms of number of bits used to store each intensity level

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

Intensity Level Resolution

256 grey levels (8 bits per pixel) 128 grey levels (7 bpp) 64 grey levels (6 bpp) 32 grey levels (5 bpp)

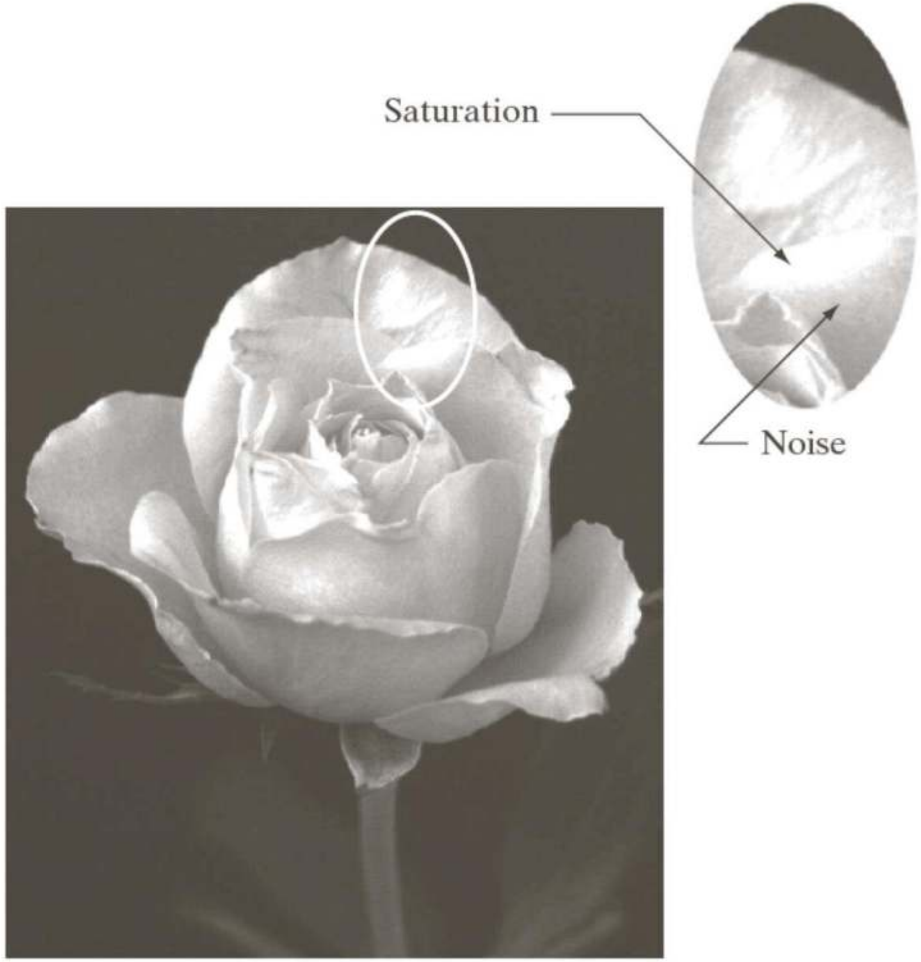


16 grey levels (4 bpp) 8 grey levels (3 bpp) 4 grey levels (2 bpp) 2 grey levels (1 bpp)

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Saturation & Noise

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Saturation: highest intensity value above which color is washed out

Noise: grainy texture pattern

Resolution: How Much Is Enough?

- The big question with resolution is always *how much is enough*?
- Depends on what is in the image *(details)* and what you would like to do with it *(applications)*
- Key questions:
  - Does image look aesthetically pleasing?
  - Can you see what you need to see in image?

Resolution: How Much Is Enough?

- Example: Picture on right okay for counting number of cars, but not for reading the number plate

Intensity Level Resolution

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Low Detail

Medium Detail

High Detail

Image File Formats

- Hundreds of image file formats. Examples
  - Tagged Image File Format (TIFF)
  - Graphics Interchange Format (GIF)
  - Portable Network Graphics (PNG)
  - JPEG, BMP, Portable Bitmap Format (PBM), etc
- Image pixel values can be
  - Grayscale: 0 – 255 range
  - Binary: 0 or 1
  - Color: RGB colors in 0-255 range (or other color model)
  - Application specific (e.g. floating point values in astronomy)

How many Bits Per Image Element?

Grayscale (Intensity Images):

Chan.	*Bits/Pix.*	*Range*	*Use*
1	1	$0 \ldots 1$	Binary image: document, illustration, fax
1	8	$0 \ldots 255$	Universal: photo, scan, print
1	12	$0 \ldots 4095$	High quality: photo, scan, print
1	14	$0 \ldots 16383$	Professional: photo, scan, print
1	16	$0 \ldots 65535$	Highest quality: medicine, astronomy

Color Images:

Chan.	*Bits/Pix.*	*Range*	*Use*
3	24	$[0 \ldots 255]^3$	RGB, universal: photo, scan, print
3	36	$[0 \ldots 4095]^3$	RGB, high quality: photo, scan, print
3	42	$[0 \ldots 16383]^3$	RGB, professional: photo, scan, print
4	32	$[0 \ldots 255]^4$	CMYK, digital prepress

Special Images:

Chan.	*Bits/Pix.*	*Range*	*Use*
1	16	$-32768 \ldots 32767$	Whole numbers pos./neg., increased range
1	32	$\pm 3.4 \cdot 10^{38}$	Floating point: medicine, astronomy
1	64	$\pm 1.8 \cdot 10^{308}$	Floating point: internal processing

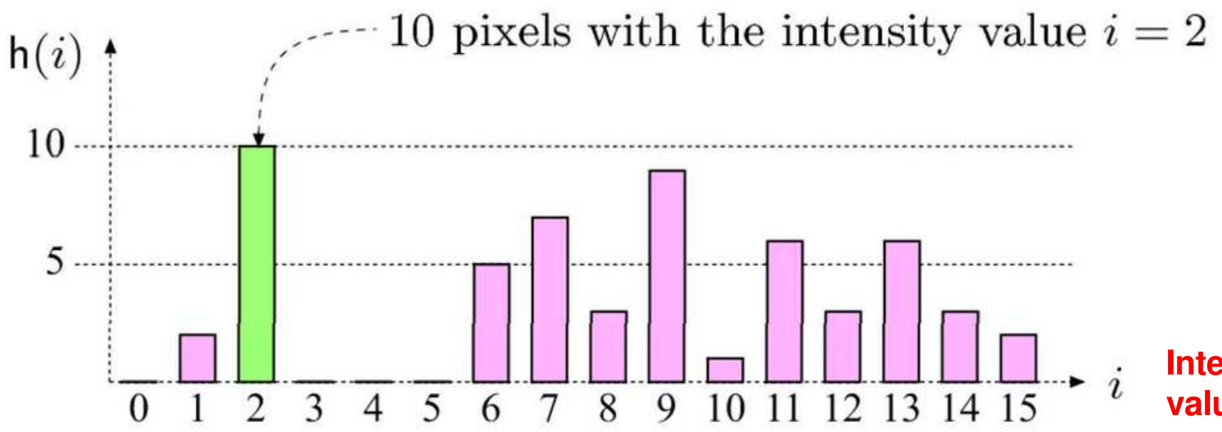
Histograms

- Histograms plots how many times (frequency) each intensity value in image occurs
- Example:
  - Image (left) has 256 distinct gray levels (8 bits)
  - Histogram (right) shows frequency (how many times) each gray level occurs

Histograms

- Many cameras display real time histograms of scene
- Helps avoid taking over-exposed pictures
- Also easier to detect types of processing previously applied to image

Histograms



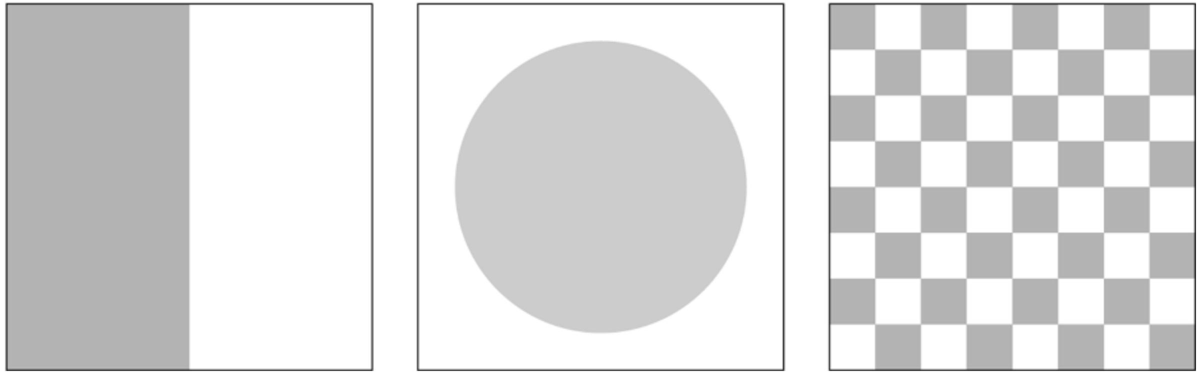
Intensity values

h(i)	0	2	10	0	0	0	5	7	3	9	1	6	3	6	3	2
i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

- E.g. K = 16, 10 pixels have intensity value = 2
- Histograms: only statistical information
- No indication of location of pixels

Histograms

- Different images can have same histogram
- 3 images below have same histogram



- Half of pixels are gray, half are white
  - Same histogram = same statisics
  - Distribution of intensities could be different
- Can we reconstruct image from histogram? No!

Histograms

- So, a histogram for a grayscale image with intensity values in range

$$I(u,v) \in [0, K-1]$$

would contain exactly K entries

- E.g. 8-bit grayscale image, $K = 2^8 = 256$
- Each histogram entry is defined as:

  h(i) = number of pixels with intensity I for all $0 < i < K$.

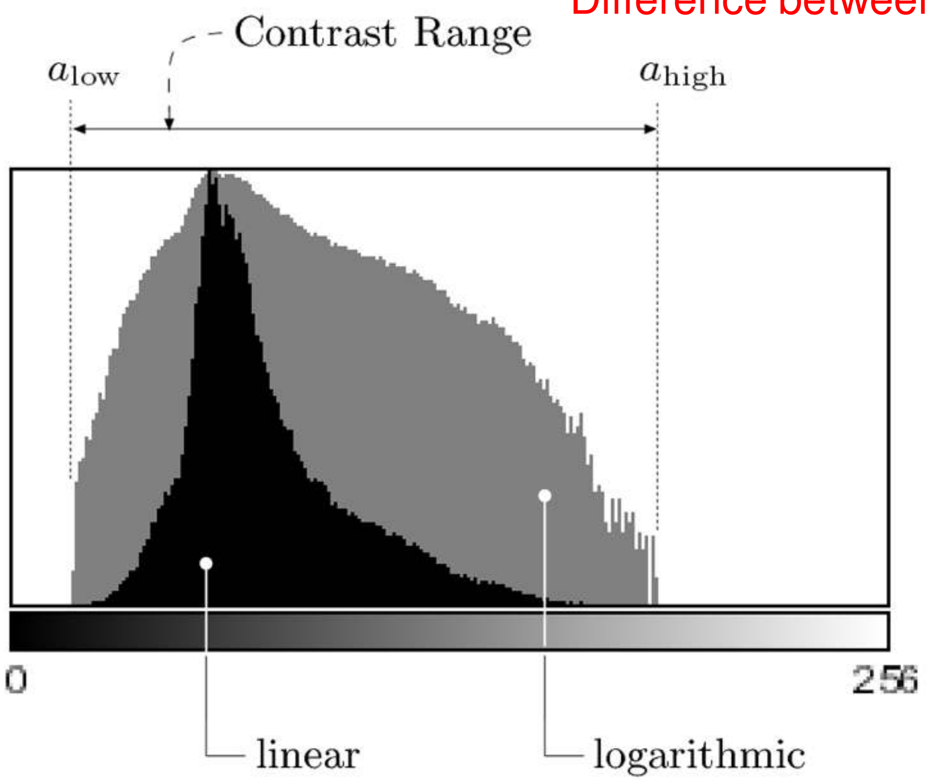
- E.g: h(255) = number of pixels with intensity = 255
- Formal definition

$$\mathsf{h}(i) = \underbrace{\mathrm{card}}_{\text{Number (size of set) of pixels}}\{(u,v) \underbrace{\mid}_{\text{such that}} I(u,v) = i\}$$

Interpreting Histograms

- Log scale makes low values more visible

Difference between darkest and lightest



Histograms

- Histograms help detect image acquisition issues
- Problems with image can be identified on histogram
  - Over and under exposure
  - Brightness
  - Contrast
  - Dynamic Range
- Point operations can be used to alter histogram. E.g
  - Addition
  - Multiplication
  - Exp and Log
  - Intensity Windowing (Contrast Modification)

Image Brightness

- Brightness of a grayscale image is the average intensity of all pixels in image

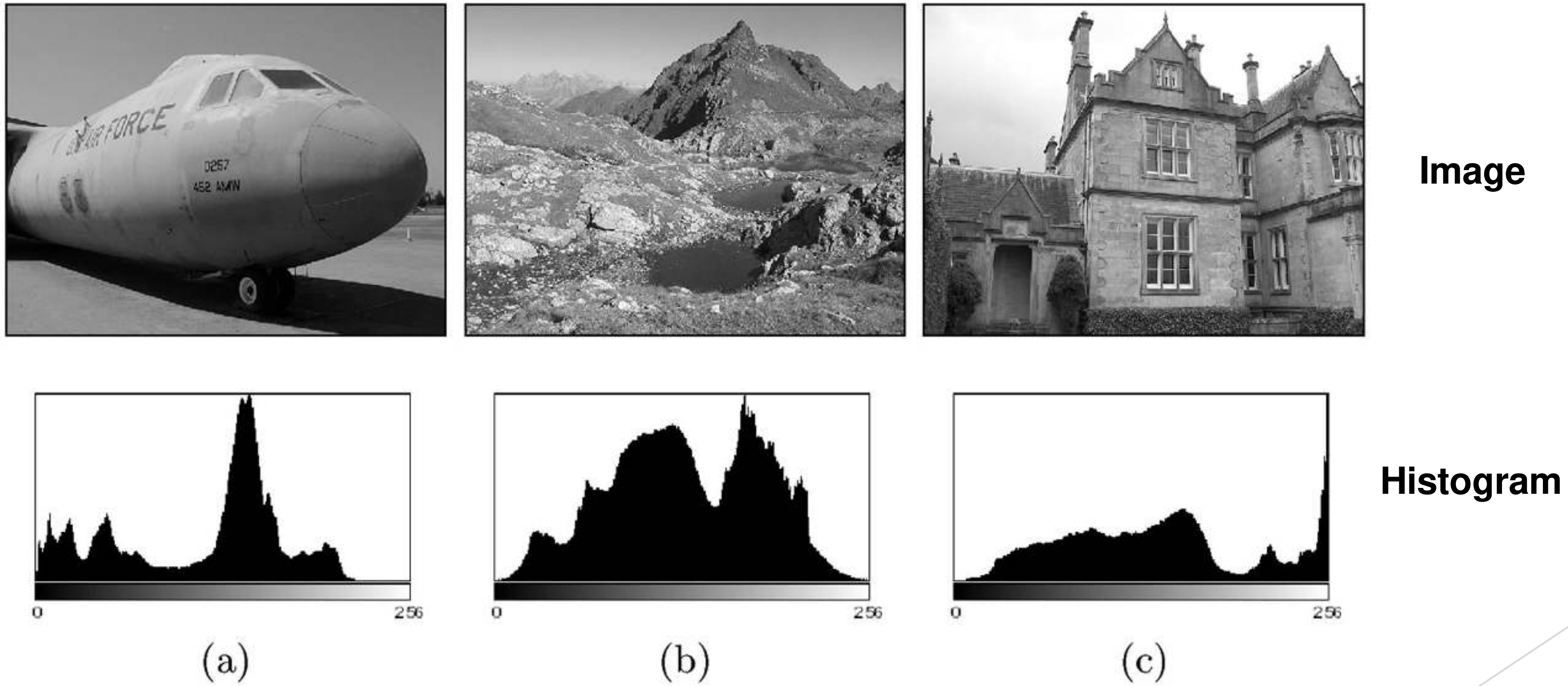
$$B(I) = \frac{1}{wh} \sum_{v=1}^{h} \sum_{u=1}^{w} I(u,v)$$

2. Divide by total number of pixels

1. Sum up all pixel intensities

Detecting Bad Exposure using Histograms

Exposure? Are intensity values spread (good) out or bunched up (bad)



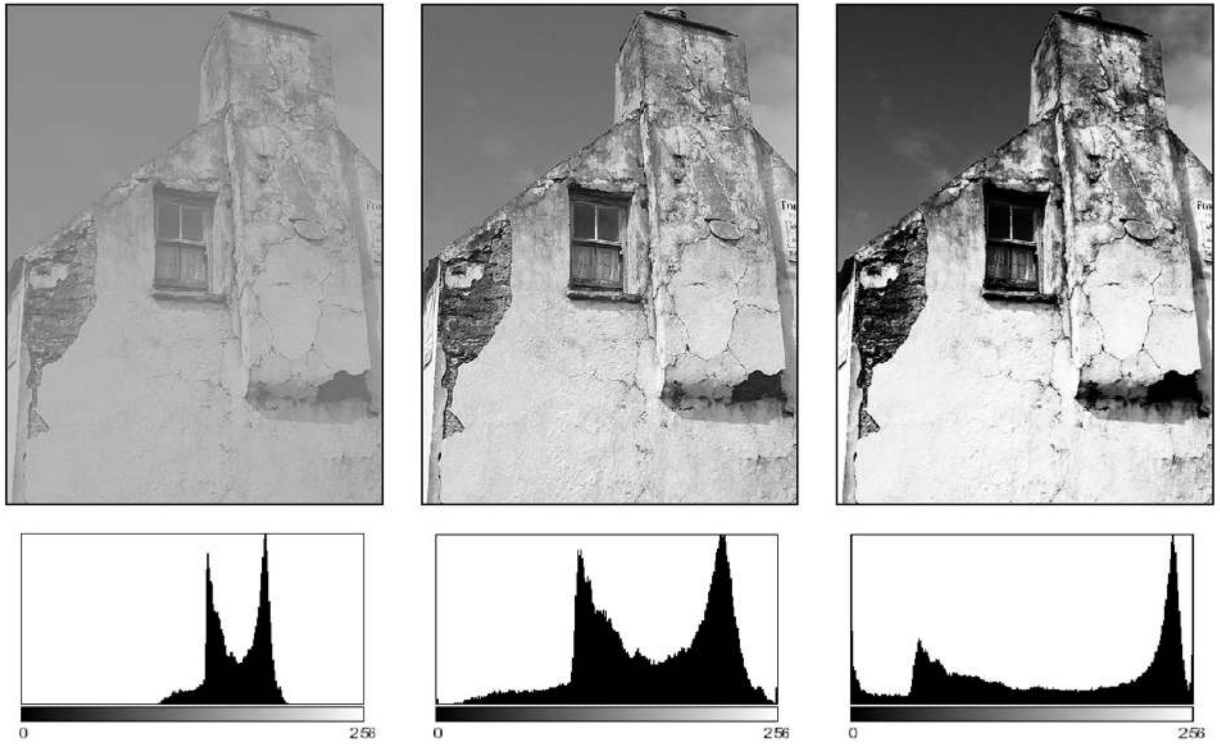
Underexposed **Properly Exposed** **Overexposed**

Image Contrast

- The contrast of a grayscale image indicates how easily objects in the image can be distinguished
- High contrast image: many distinct intensity values
- Low contrast: image uses few intensity values

Histograms and Contrast

Good Contrast? Widely spread intensity values
+ large difference between min and max intensity values



Image

Histogram

(a) (b) (c)

Low contrast **Normal contrast** **High contrast**

Contrast Equation?

- Many different equations for contrast exist
- Examples:

$$\text{Contrast} = \frac{\text{Change in Luminance}}{\text{Average Luminance}}$$

- Michalson's equation for contrast

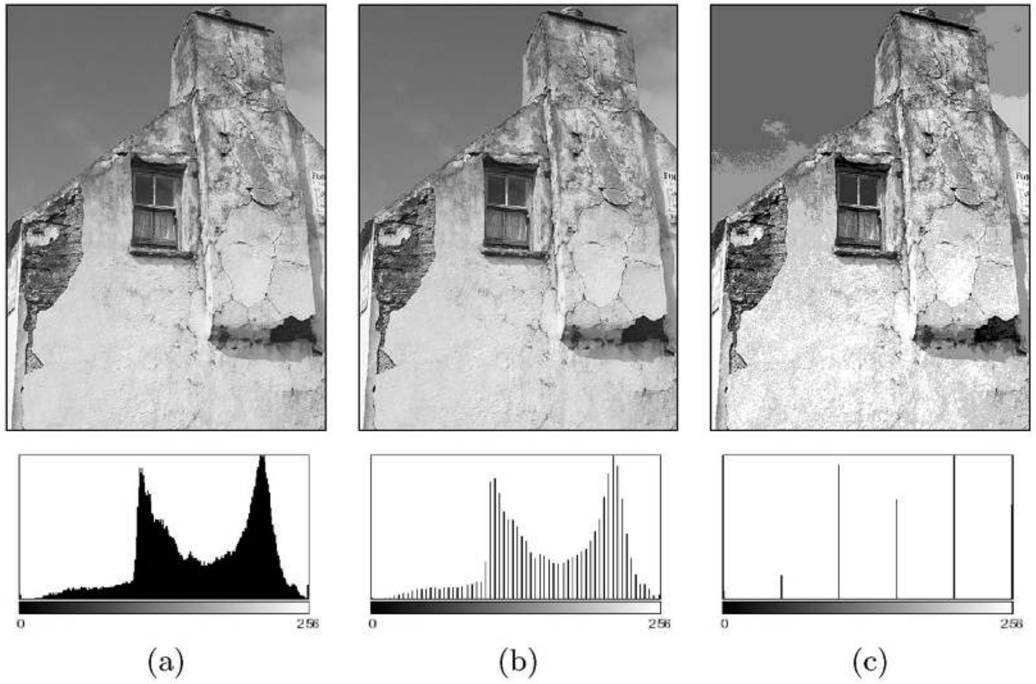
$$C_M(I) = \frac{\max(I) - \min(I)}{\max(I) + \min(I)}$$

Contrast Equation?

- These equations work well for simple images with 2 luminances (i.e. uniform foreground and background)
- Does not work well for complex scenes with many luminances or if min and max intensities are small

Histograms and Dynamic Range

- Dynamic Range: Number of distinct pixels in image



High Dynamic Range

Low Dynamic Range (64 intensities)

Extremely low Dynamic Range (6 intensity values)

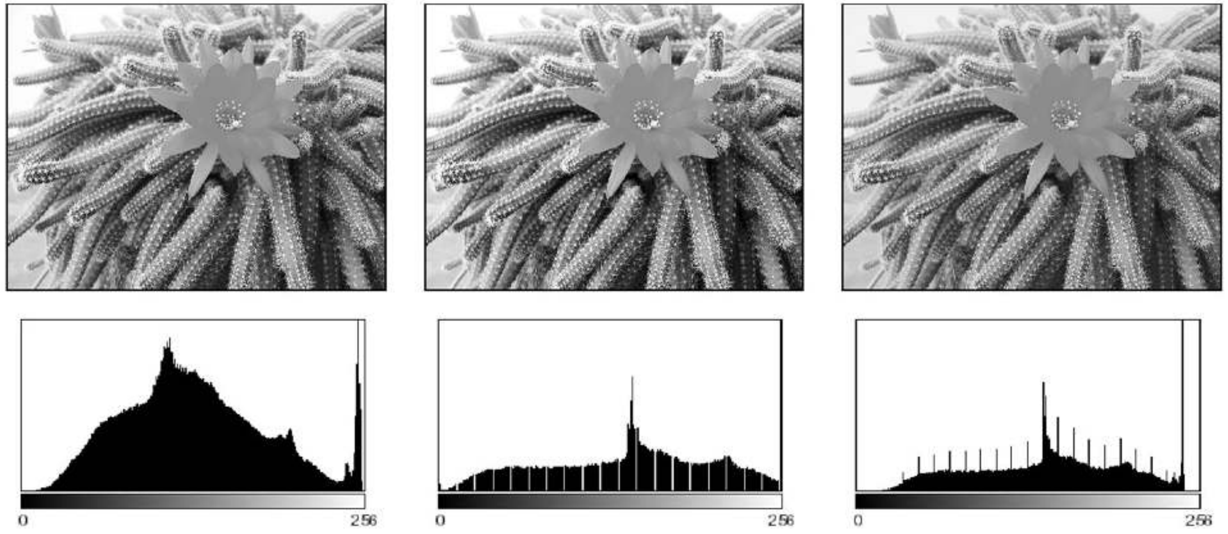
- Difficult to increase image dynamic range (e.g. interpolation)
- HDR (12-14 bits) capture typical, then down-sample

High Dynamic Range Imaging

- High dynamic range means very bright and very dark parts in a single image (many distinct values)
- Dynamic range in photographed scene may exceed number of available bits to represent pixels
- Solution:
  - Capture multiple images at different exposures
  - Combine them using image processing

Detecting Image Defects using Histograms

- No “best” histogram shape, depends on application
- Image defects
  - Saturation: scene illumination values outside the sensor’s range are set to its min or max values => results in spike at ends of histogram
  - Spikes and Gaps in manipulated images (not original). Why?

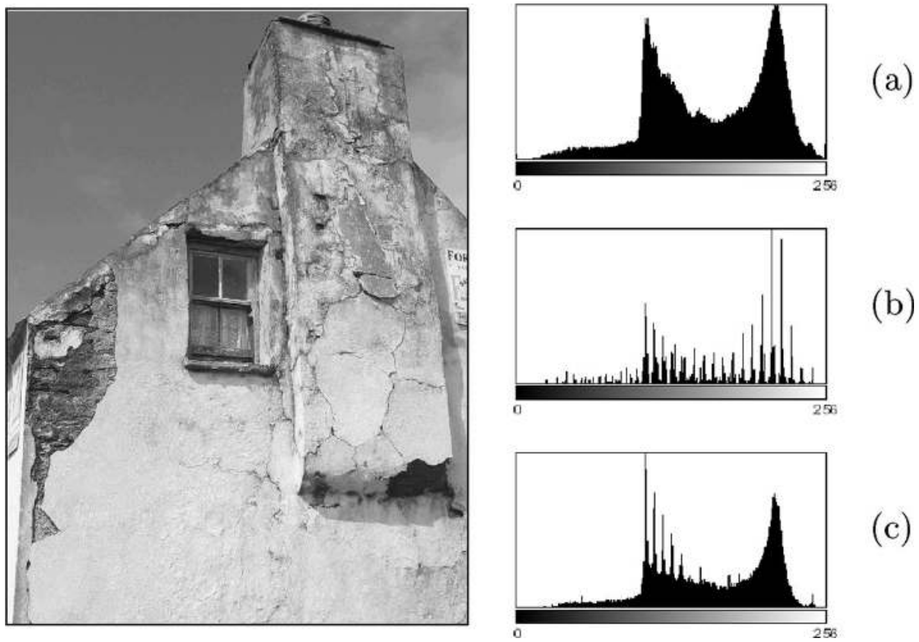


(a) (b) (c)

Image Defects: Effect of Image Compression

- Histograms show impact of image compression
- Example: in GIF compression, dynamic range is reduced to only few intensities (quantization)

Original Image



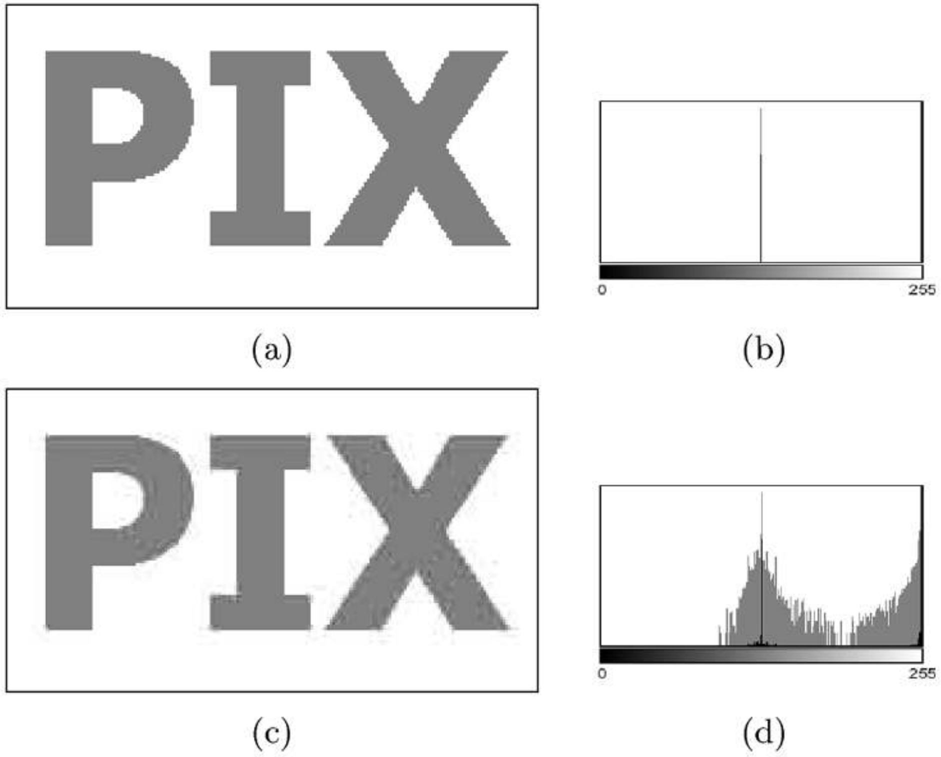
Original Histogram

Histogram after GIF conversion

Fix? Scaling image by 50% and Interpolating values recreates some lost colors
But GIF artifacts still visible

Effect of Image Compression

- Example: Effect of JPEG compression on line graphics
- JPEG compression designed for color images



(a) (b) (c) (d)

Original histogram has only 2 intensities (gray and white)

JPEG image appears dirty, fuzzy and blurred

Its Histogram contains gray values not in original

Computing Histograms

```java
public class Compute_Histogram implements PlugInFilter {

    public int setup(String arg, ImagePlus img) {
        return DOES_8G + NO_CHANGES;
    }

    public void run(ImageProcessor ip) {
        int[] H = new int[256]; // histogram array
        int w = ip.getWidth();
        int h = ip.getHeight();

        for (int v = 0; v < h; v++) {
            for (int u = 0; u < w; u++) {
                int i = ip.getPixel(u,v);
                H[i] = H[i] + 1;
            }
        }
        ... //histogram H[] can now be used
    }

} // end of class Compute_Histogram
```

Receives 8-bit image, Will not change it

Create array to store histogram computed

Get width and height of image

Iterate through image pixels, add each intensity to appropriate histogram bin

ImageJ Histogram Function

- ImageJ has a histogram function (`getHistogram( )`)
- Prior program can be simplified if we use it

```
public void run(ImageProcessor ip) {
    int[] H = ip.getHistogram();
    ... // histogram H[] can now be used
}
```

Returns histogram as an array of integers

Large Histograms: Binning

- High resolution image can yield very large histogram
- Example: 32-bit image = 2^{32} = 4,294,967,296 columns
- Such a large histogram impractical to display
- Solution? Binning!
  - Combine ranges of intensity values into histogram columns

So, given the image $I : \Omega \to [0, K-1]$, the binned histogram for I is the function

$$h(i) = \text{card}\{(u, v) | \ a_i \leq I(u, v) < a_{i+1}\},$$

where $0 = a_0 < a_1 < \ldots < a_B = K$.

Number (size of set) of pixels

such that

Pixel's intensity is between a_i and a_{i+1}

Calculating Bin Size

- Typically use equal sized bins
- Bin size? $\dfrac{\textit{Number of distinct values in image}}{\textit{Number of bins}}$

- Example: To create 256 bins from 14-bit image

$$\textit{Bin size} = \frac{2^{14}}{256} = 64$$

$$\begin{array}{lcrcl}
\mathsf{h}(0) & \leftarrow & 0 \leq I(u,v) < & 64 \\
\mathsf{h}(1) & \leftarrow & 64 \leq I(u,v) < & 128 \\
\mathsf{h}(2) & \leftarrow & 128 \leq I(u,v) < & 192 \\
\vdots & & \vdots \quad \vdots & \vdots \\
\mathsf{h}(j) & \leftarrow & a_j \leq I(u,v) < & a_{j+1} \\
\vdots & & \vdots \quad \vdots & \vdots \\
\mathsf{h}(255) & \leftarrow & 16320 \leq I(u,v) < & 16384
\end{array}$$

Binned Histogram

- To calculate which bin a pixel's intensity belongs to

$$\frac{I(u,v)}{k_B} = \frac{I(u,v)}{K/B} = I(u,v) \cdot \frac{B}{K}$$

- Previous example, B = 256, K = 2^{14} = 16384

```
int[] binnedHistogram(ImageProcessor ip) {
  int K = 256; // number of intensity values
  int B = 32; // size of histogram, must be defined
  int[] H = new int[B]; // histogram array
  int w = ip.getWidth();
  int h = ip.getHeight();

  for (int v = 0; v < h; v++) {
    for (int u = 0; u < w; u++) {
      int a = ip.getPixel(u, v);
      int i = a * B / K; // integer operations only!
      H[i] = H[i] + 1;
    }
  }
  // return binned histogram
  return H;
}
```

Create array to store histogram computed

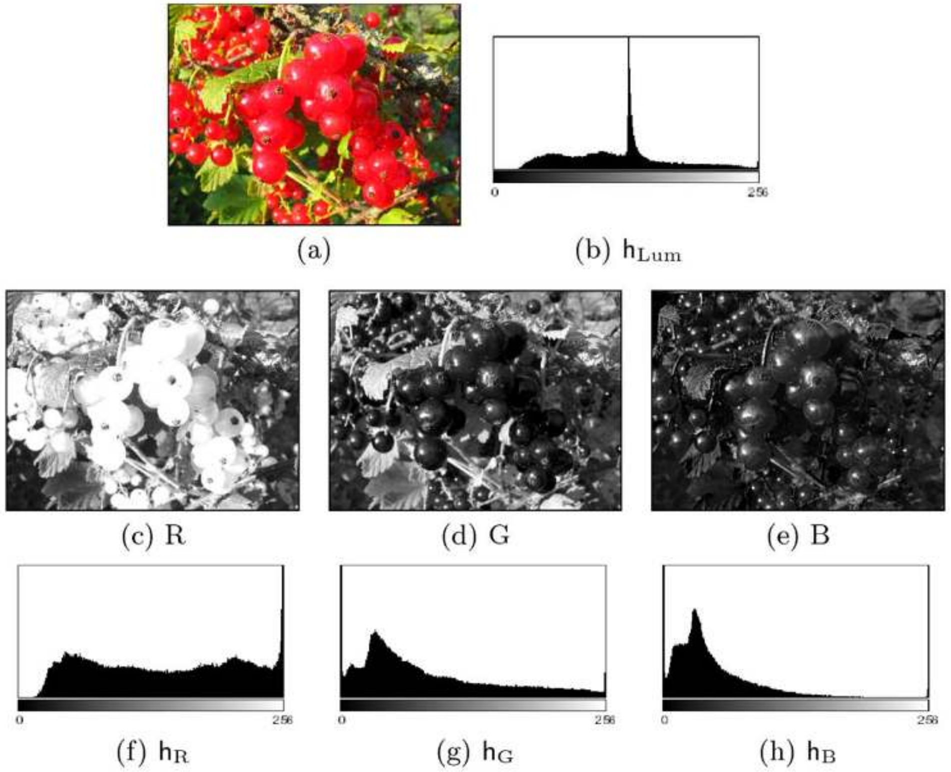
Calculate which bin to add pixel's intensity

Increment corresponding histogram

Color Image Histograms

Two types:

1. Intensity histogram:
   - Convert color image to gray scale
   - Display histogram of gray scale
2. Individual Color Channel Histograms: 3 histograms (R,G,B)



Color Image Histograms

- Both types of histograms provide useful information about lighting, contrast, dynamic range and saturation effects
- No information about the actual color distribution!
- Images with totally different RGB colors can have same R, G and B histograms
- Solution to this ambiguity is the Combined Color Histogram.
    - More on this later

Cumulative Histogram

- Useful for certain operations (e.g. histogram equalization) later
- Analogous to the Cumulative Density Function (CDF)
- Definition:

$$\mathsf{H}(i) = \sum_{j=0}^{i} \mathsf{h}(j) \quad \text{for} \;\; 0 \le i < K$$

- Recursive definition

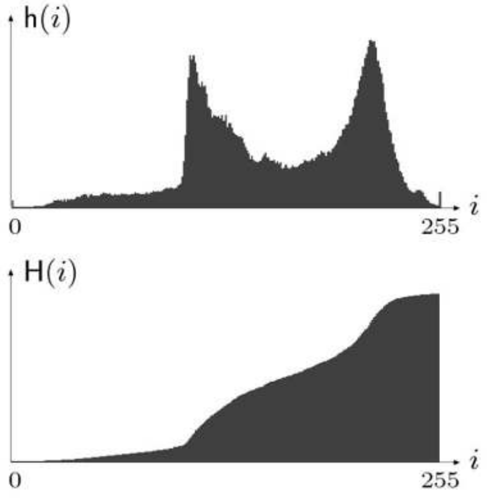
$$\mathsf{H}(i) = \begin{cases} \mathsf{h}(0) & \text{for} \;\; i = 0 \\ \mathsf{H}(i-1) + \mathsf{h}(i) & \text{for} \;\; 0 < i < K \end{cases}$$

- Monotonically increasing

$$\mathsf{H}(K-1) = \sum_{j=0}^{K-1} \mathsf{h}(j) = M \cdot N$$

Last entry of Cum. histogram

Total number of pixels in image



Point Operations

- Point operations changes a pixel's intensity value according to some function (don't care about pixel's neighbor)

$$a' \leftarrow f(a)$$
$$I'(u,v) \leftarrow f\big(I(u,v)\big)$$

- Also called a homogeneous operation
- New pixel intensity depends on
  - Pixel's previous intensity *I(u,v)*
  - Mapping function *f*( )
- Does not depend on
  - Pixel's location *(u,v)*
  - Intensities of neighboring pixels

Some Homogeneous Point Operations

- Addition (Changes brightness)

$$f(p) = p + k \qquad \text{E.g.} \quad f_{\text{bright}}(p) = p + 10$$

- Multiplication (Stretches/shrinks image contrast range)

$$f(p) = k \times p \qquad \text{E.g.} \quad f_{\text{contrast}}(p) = p \text{ x } 1.5$$

- Real-valued functions

$$\exp(x), \log(x), (1/x), x^k, \text{etc.}$$

- Quantizing pixel values
- Global thresholding
- Gamma correction

Point Operation Pseudocode

- Input: Image with pixel intensities *I(u,v)* defined on [1 .. *w*] x [1 .. *H*]
- Output: Image with pixel intensities *I'(u,v)*

```
for v = 1 .. h
    for u = 1 .. w
        set I(u, v) = f (I(u,v))
```

Non-Homogeneous Point Operation

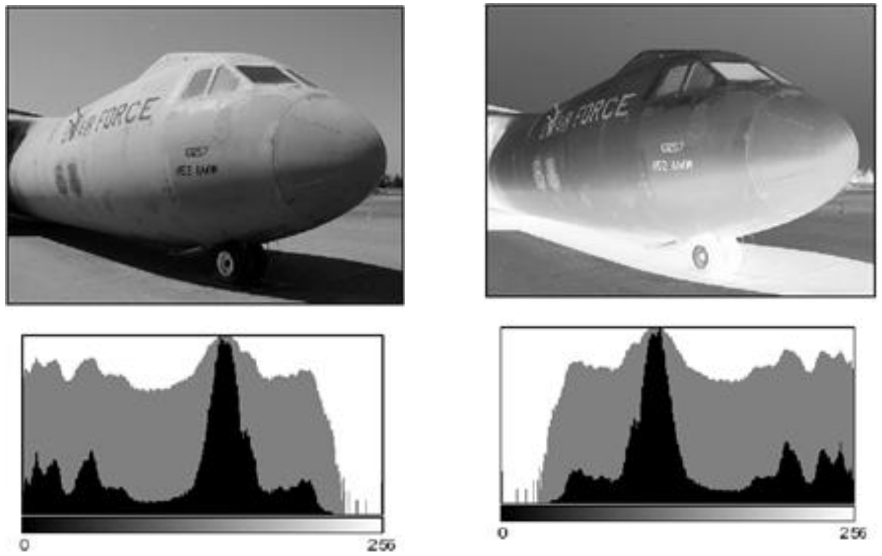
- New pixel value depends on:
  - Old value + pixel's location *(u,v)*

$$a' \leftarrow g(a, u, v)$$
$$I'(u, v) \leftarrow g\left(I(u, v), u, v\right)$$

Inverting Images

$$f_{\text{invert}}(a) = -a + a_{\max} = a_{\max} - a$$

- 2 steps
  1. Multiple intensity by -1
  2. Add constant (e.g. a_{max}) to put result in range [0,a_{max}]
- Implemented as ImageJ method `invert( )`



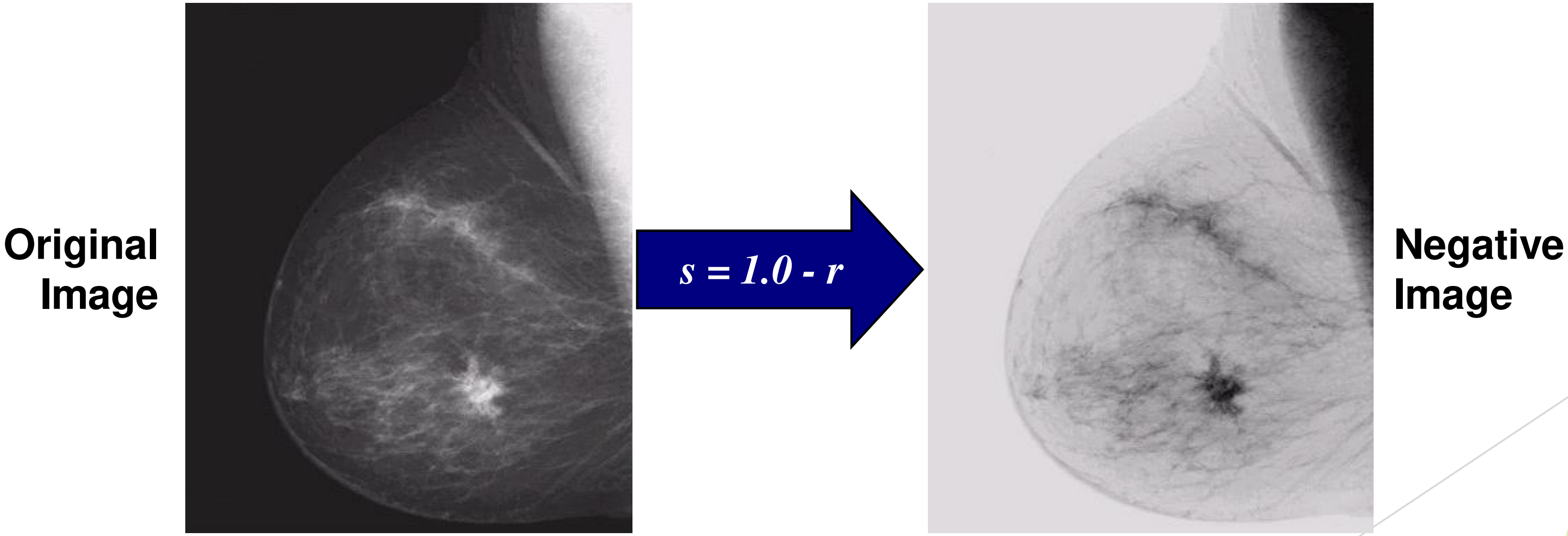
(a) Original

(c) Inverted Image

Image Negatives (Inverted Images)

- Image negatives useful for enhancing white or grey detail embedded in dark regions of an image
  - Note how much clearer the tissue is in the negative image of the mammogram below

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Thresholding

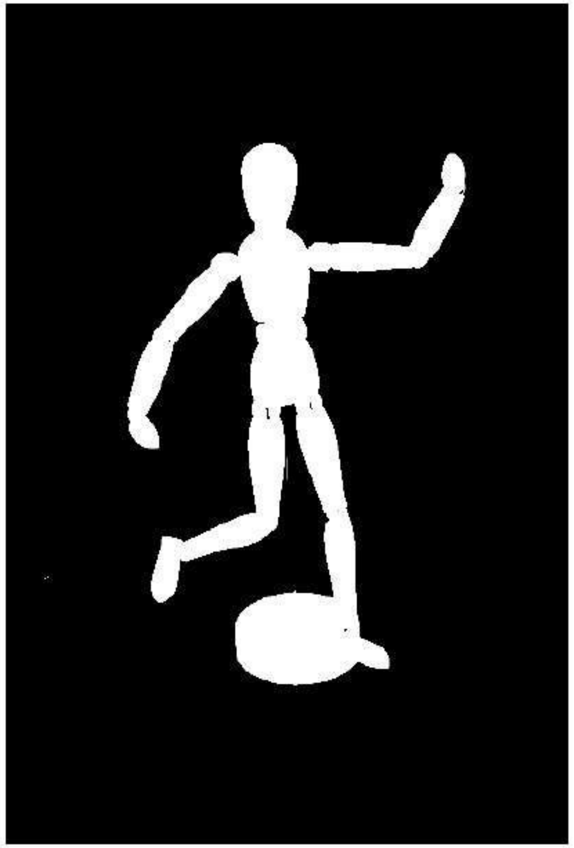
- Input values below **threshold** a_{th} set to a_0
- Input values above **threshold** a_{th} set to a_1

$$f_{\text{threshold}}(a) = \begin{cases} a_0 & \text{for } \; a < a_{\text{th}} \\ a_1 & \text{for } \; a \geq a_{\text{th}} \end{cases}$$

- Converts grayscale image to binary image (binarization) if
  - $a_0 = 0$
  - $a_1 = 1$
- Implemented as imageJ method `threshold( )`

Thresholding Example

Original Image

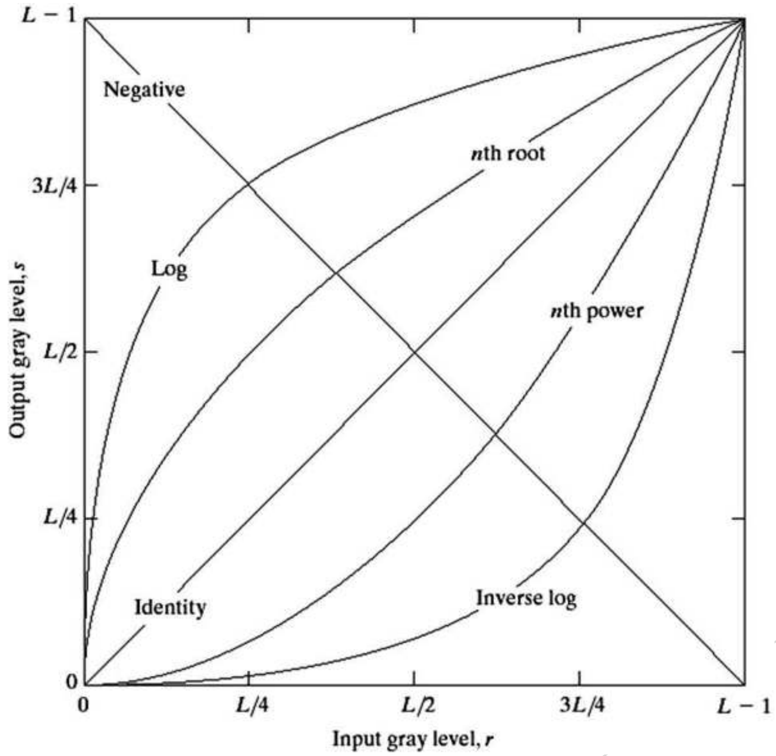


Thresholded Image

Basic Grey Level Transformations

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

- 3 most common gray level transformation:
  - Linear
    - Negative/Identity
  - Logarithmic
    - Log/Inverse log
  - Power law
    - n^{th} power/n^{th} root

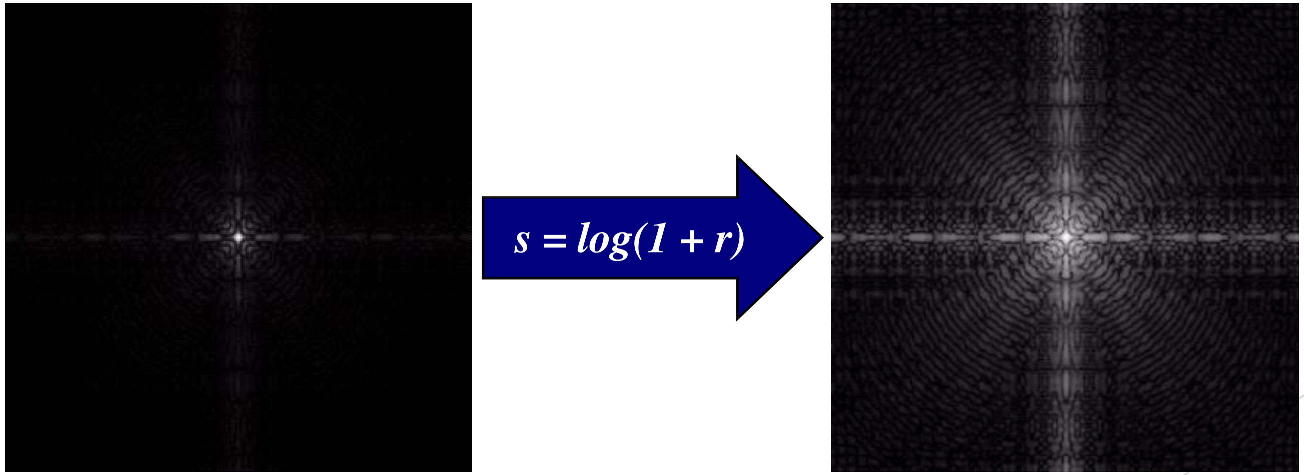


Logarithmic Transformations

- Maps narrow range of input levels => wider range of output values
- Inverse log transformation does opposite transformation
- The general form of the log transformation is

New pixel value $\longrightarrow s = c * log(1 + r) \longleftarrow$ Old pixel value

- Log transformation of Fourier transform shows more detail



Power Law Transformations

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

- Power law transformations have the form

$$s = c * r^{\gamma}$$

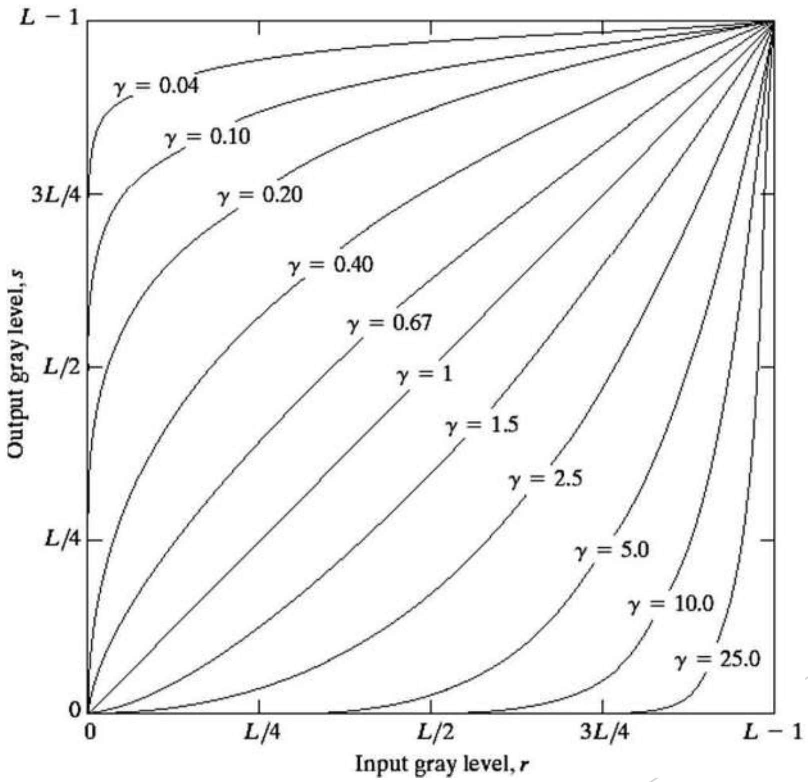
Power

New pixel value

Constant

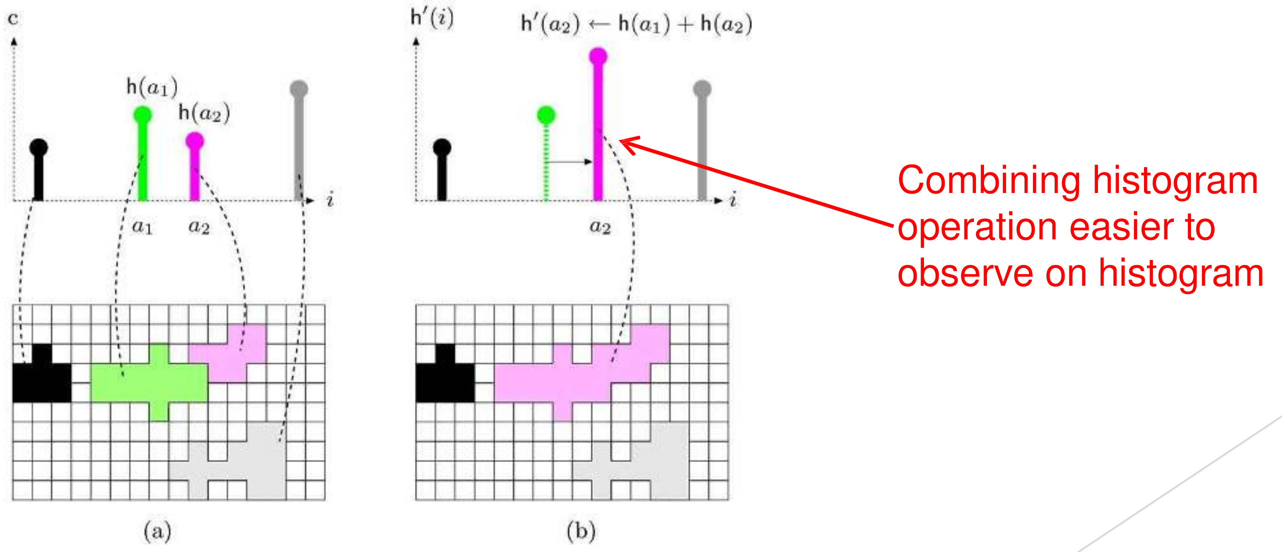
Old pixel value

- Map narrow range of dark input values into wider range of output values or vice versa
- Varying γ gives a whole family of curves



Point Operations and Histograms

- Effect of some point operations easier to observe on histograms
  - Increasing brightness
  - Raising contrast
  - Inverting image
- Point operations only shift, merge histogram entries
- Operations that merge histogram bins are irreversible



Automatic Contrast Adjustment

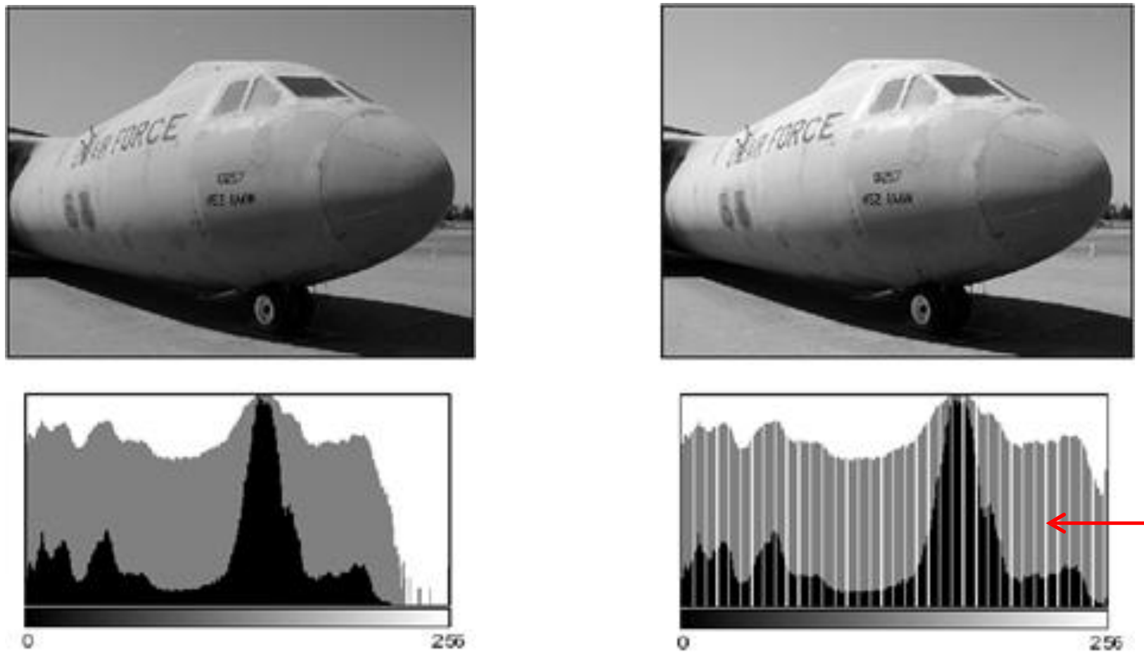
- Point operation that modifies pixel intensities such that available range of values is fully covered
- Algorithm:
  - Find high and lowest pixel intensities a_{low}, a_{high}
  - Linear stretching of intensity range

$$f_{\text{ac}}(a) = a_{\text{min}} + (a - a_{\text{low}}) \cdot \frac{a_{\text{max}} - a_{\text{min}}}{a_{\text{high}} - a_{\text{low}}}$$

If a_{min} = 0 and a_{max} = 255

$$f_{\text{ac}}(a) = (a - a_{\text{low}}) \cdot \frac{255}{a_{\text{high}} - a_{\text{low}}}$$

Effects of Automatic Contrast Adjustment



(a) (b)

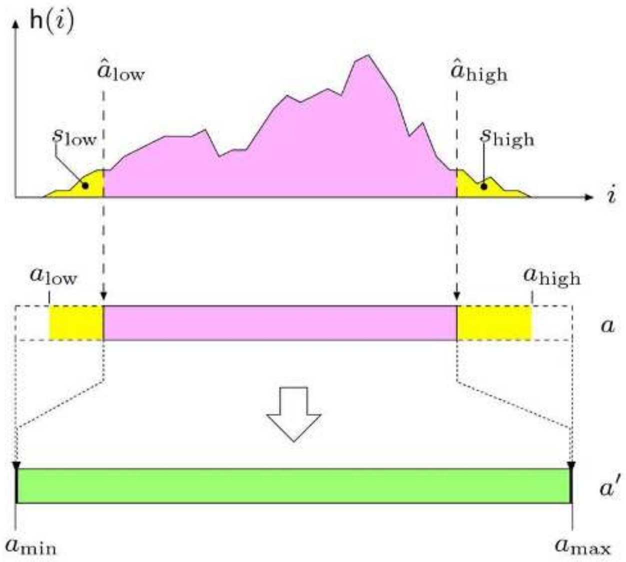
Original

Result of automatic Contrast Adjustment

Linearly stretching range causes gaps in histogram

Modified Contrast Adjustment

- Better to map only certain range of values
- Get rid of tails (usually noise) based on predefined percentiles ($s_{\text{low}}, s_{\text{high}}$)



$$\hat{a}_{\text{low}} = \min\{\, i \mid \mathsf{H}(i) \geq M \cdot N \cdot s_{\text{low}} \}$$

$$\hat{a}_{\text{high}} = \max\{\, i \mid \mathsf{H}(i) \leq M \cdot N \cdot (1 - s_{\text{high}}) \}$$

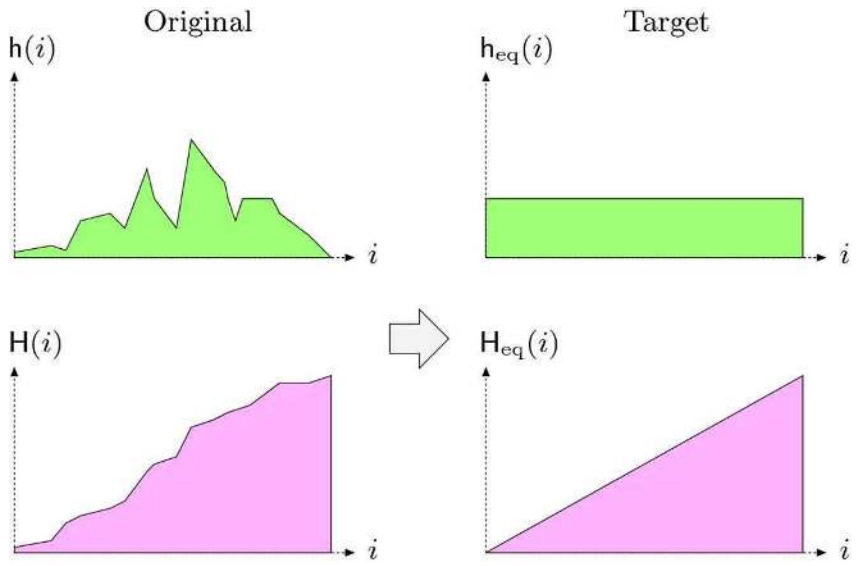
$$f_{\text{mac}}(a) = \begin{cases} a_{\text{min}} & \text{for } a \leq \hat{a}_{\text{low}} \\ a_{\text{min}} + (a - \hat{a}_{\text{low}}) \cdot \dfrac{a_{\text{max}} - a_{\text{min}}}{\hat{a}_{\text{high}} - \hat{a}_{\text{low}}} & \text{for } \hat{a}_{\text{low}} < a < \hat{a}_{\text{high}} \\ a_{\text{max}} & \text{for } a \geq \hat{a}_{\text{high}} \end{cases}$$

Histogram Equalization

- Adjust 2 different images to make their histograms (intensity distributions) similar
- Apply a point operation that changes histogram of modified image into uniform distribution

Histogram

Cumulative Histogram



Histogram Equalization

Spreading out the frequencies in an image (or equalizing the image) is a simple way to improve dark or washed out images

Can be expressed as a transformation of histogram

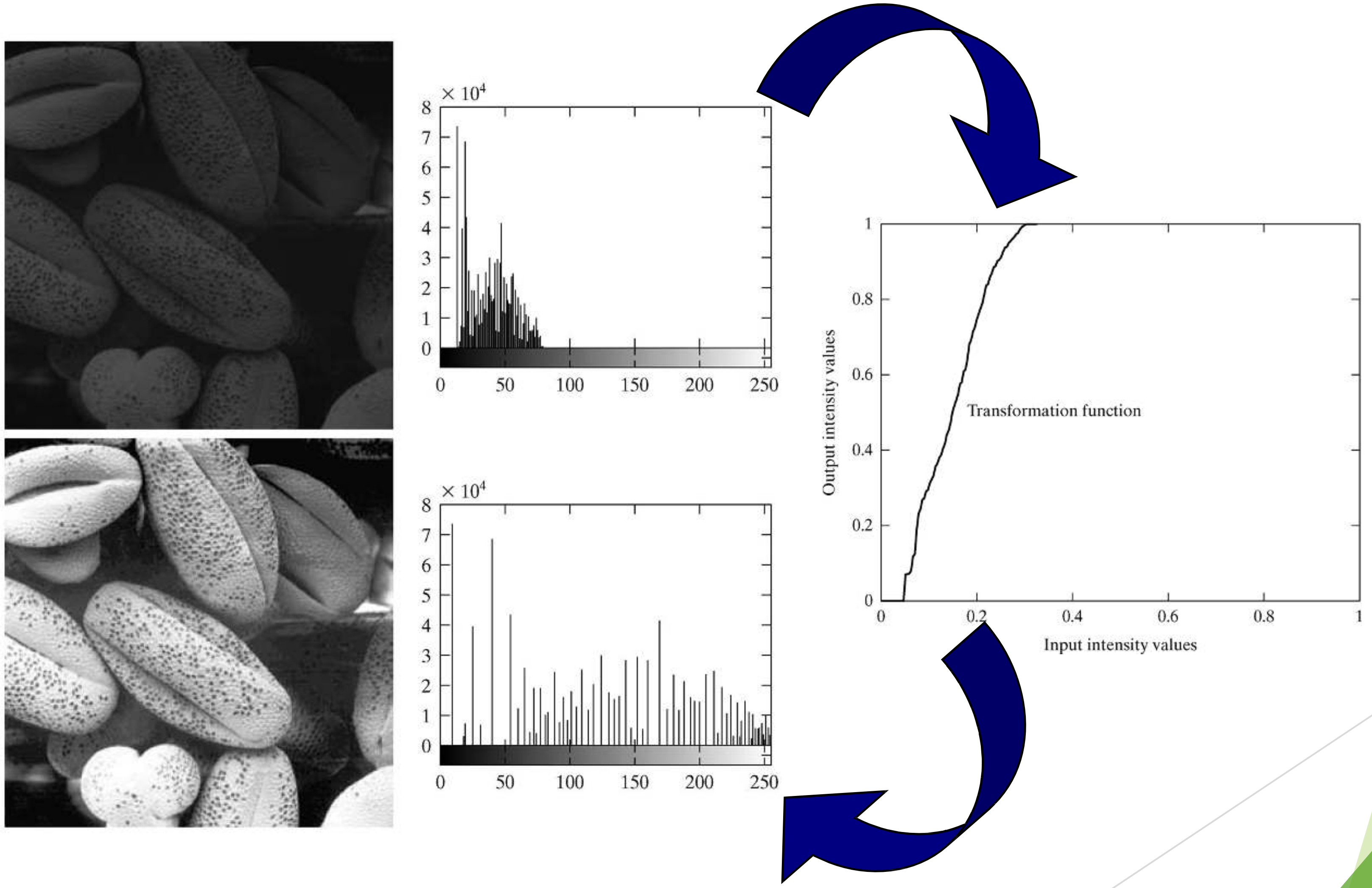
- r_k: input intensity
- s_k: processed intensity
- k: the intensity range (e.g 0.0 – 1.0)

processed intensity → $$s_k = T(r_k)$$ ← input intensity

Intensity range
(e.g 0 – 255)

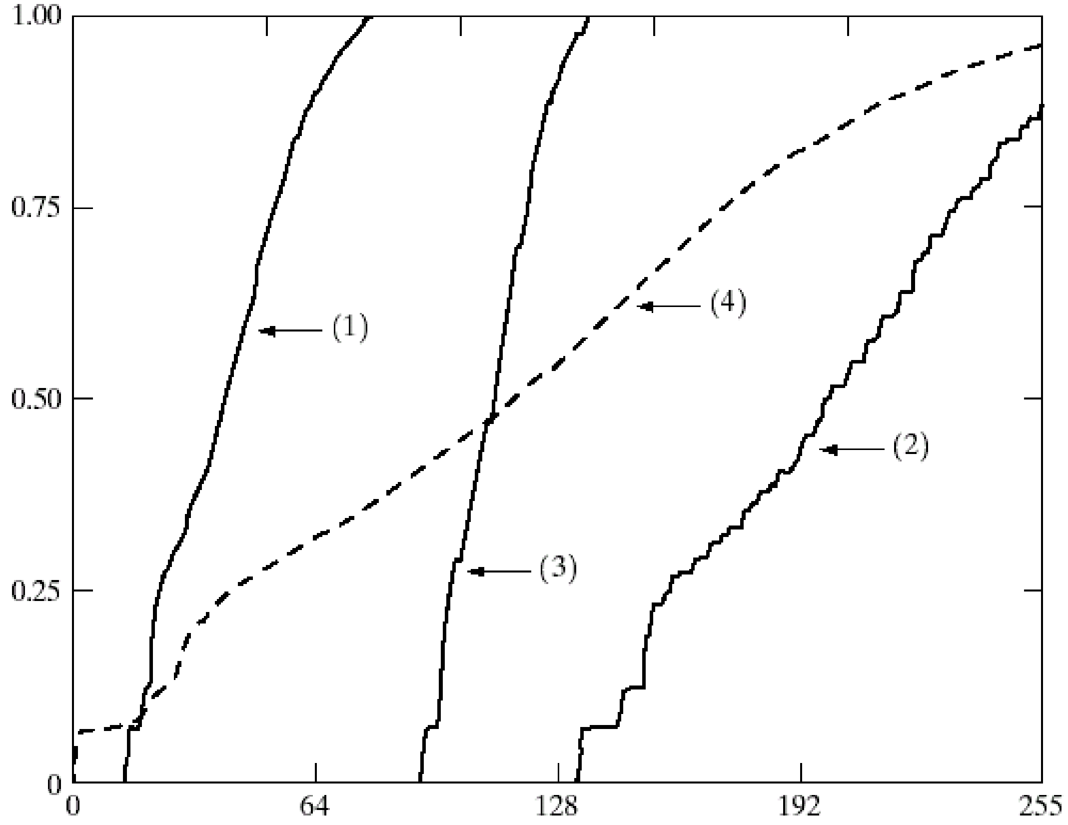
Equalization Transformation Function

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



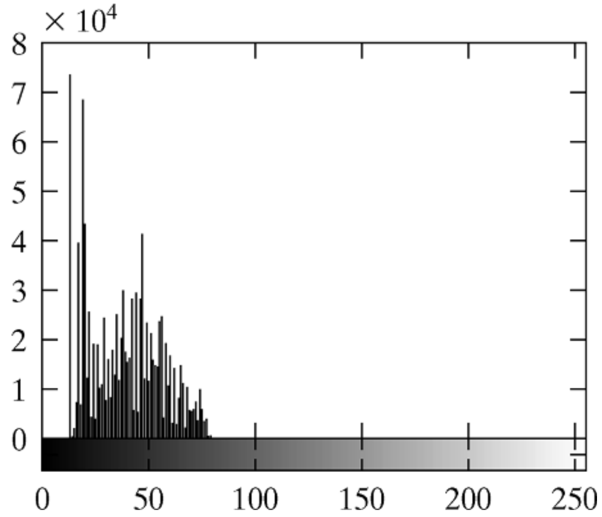
Equalization Transformation Functions

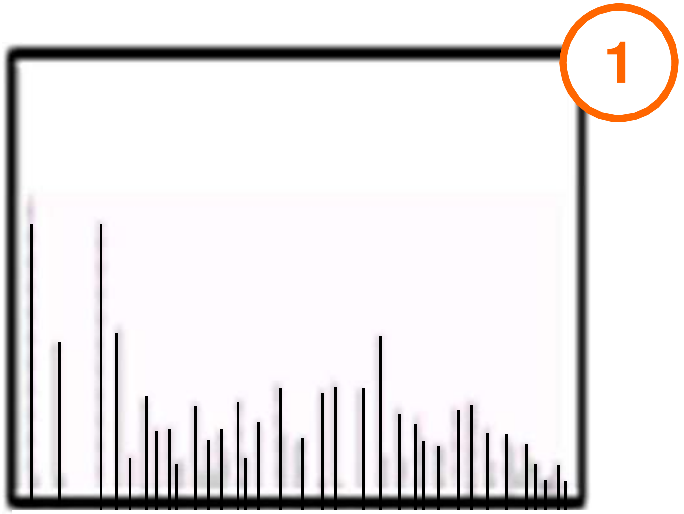
Different equalization function (1-4) may be used

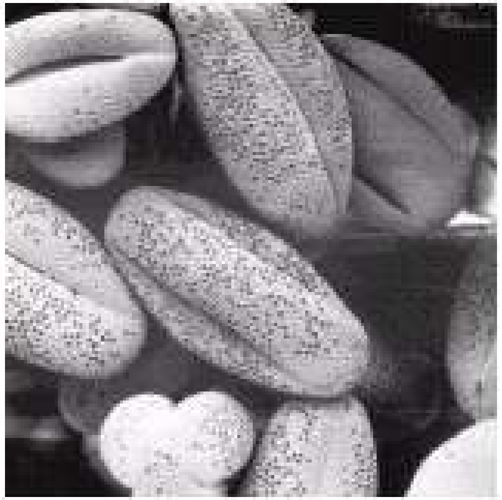


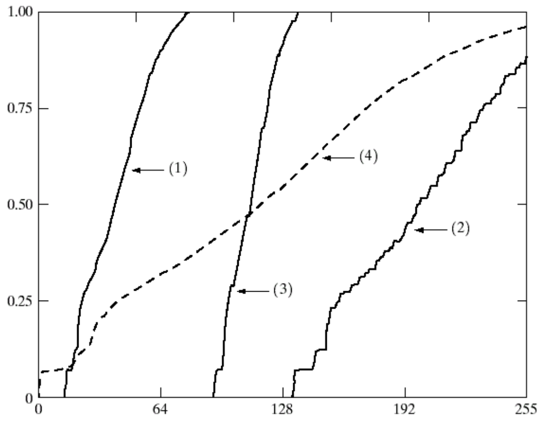
Equalization Examples

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



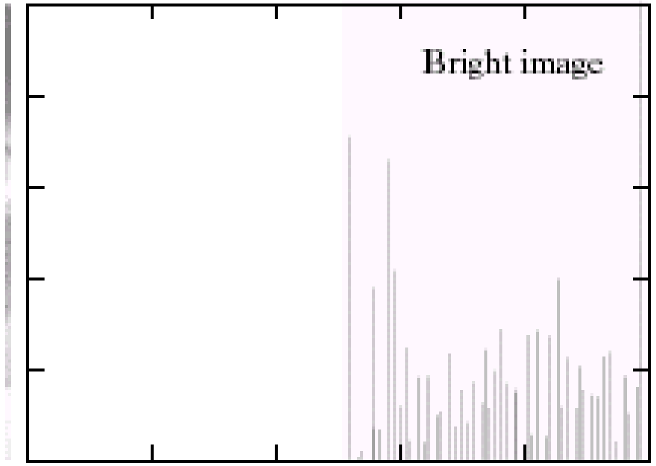




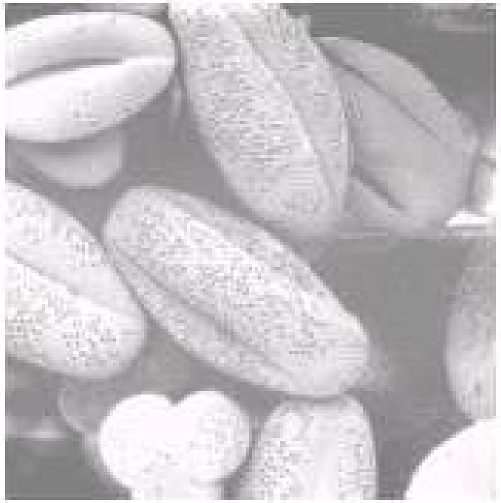


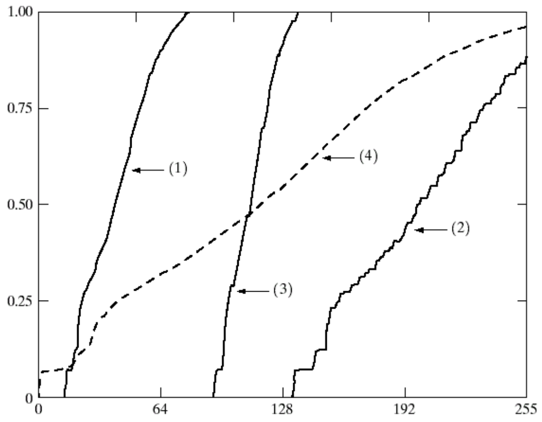
Equalization Examples

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

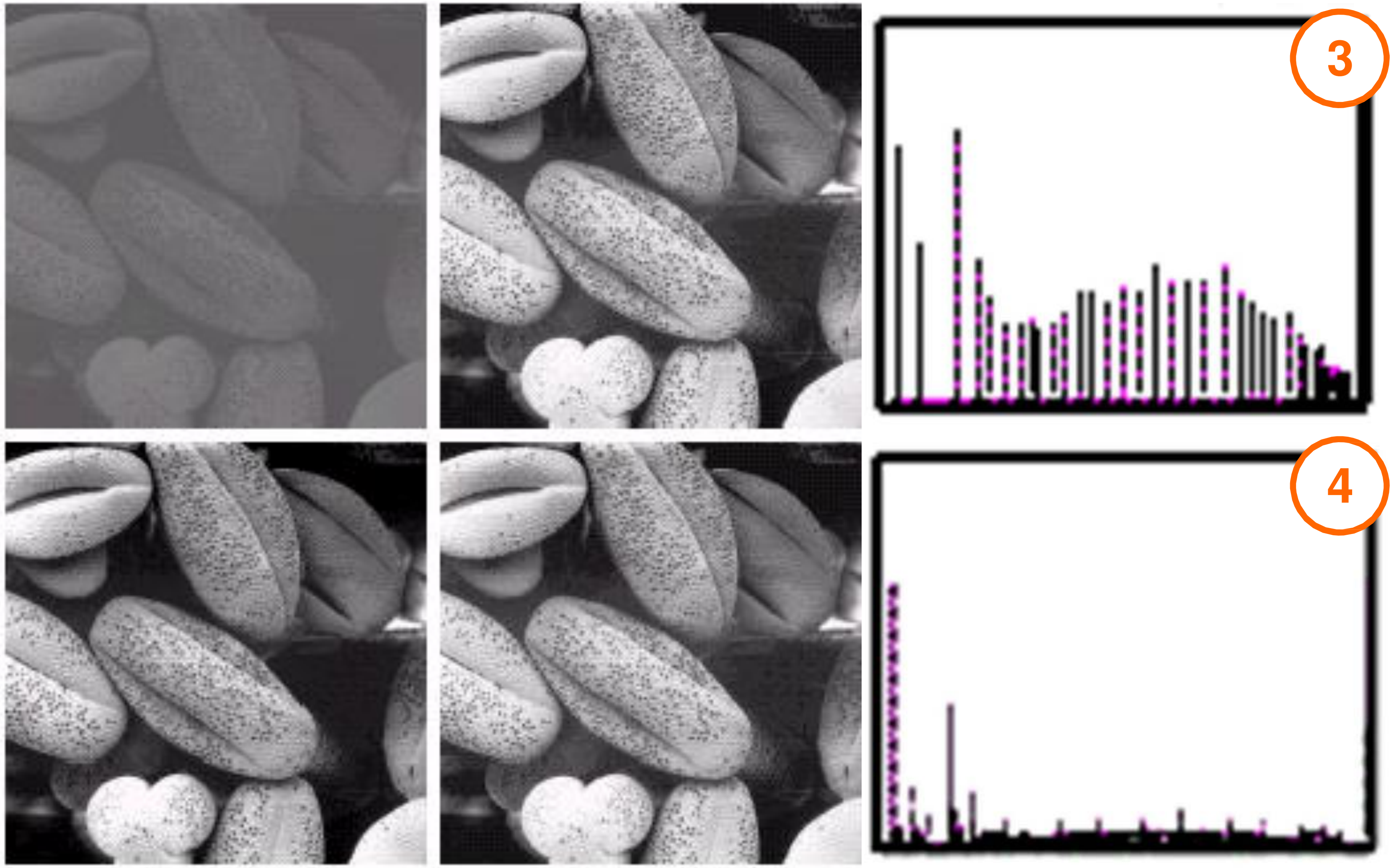


2





Equalization Examples



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

References

- Wilhelm Burger and Mark J. Burge, Digital Image Processing, Springer, 2008
  - Histograms (Ch 4)
  - Point operations (Ch 5)
- University of Utah, CS 4640: Image Processing Basics, Spring 2012
- Rutgers University, CS 334, Introduction to Imaging and Multimedia, Fall 2012
- Gonzales and Woods, Digital Image Processing (3rd edition), Prentice Hall